

Supplementary Appendix to “Belief Disagreement, Instability, and Uncertainty under Stochastic Recall”

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S.1 Existing Belief Formation Models: Detailed Discussion

This appendix provides a detailed comparison of the empirical patterns documented in Section 2 of the main paper with leading frameworks of belief formation. For ease of reference, Table S.1 and its continuation label the empirical findings of the main paper (Sections 2 and 4 to 6) as Facts 1 to 17; this numbering is used only in this appendix.

Informational frictions and dynamic consistency A key benchmark in canonical informational-friction models is dynamic consistency conditional on a fixed information set: absent new signals, beliefs do not systematically change. Rational inattention (Sims, 2003; Mackowiak and Wiederholt, 2009), sticky information (Mankiw, Reis and Wolfers, 2003), and noisy information (Lucas, 1972; Woodford, 2003) generate disagreement through heterogeneous attention, updating frequency, or imperfect signal extraction. These frameworks typically focus on differences in information acquisition or processing. Our evidence instead isolates a channel that operates even when information histories coincide and no new signals arrive. In addition, these models abstract from stochastic retrieval of stored observations and therefore do not directly address dispersion arising from variation in what is recalled from a shared history (**Fact 1**). Panel A of Table S.1 summarizes how each framework maps onto the field-study facts.

Table S.1: EMPIRICAL PATTERNS AND PREDICTIONS OF BELIEF-FORMATION FRAMEWORKS

Empirical belief pattern	Informational frictions	Salience / diagnostic models	Stochastic recall
<i>Panel A. Field studies 1–4 (Section 2 of the main paper)</i>			
Fact 1: Heterogeneous recall of identical information histories	Abstracts from retrieval heterogeneity under shared information	Retrieval effectively deterministic under many memory samples	Direct implication
Fact 2: Recalled information predicts beliefs	Consistent with framework	Consistent with framework	Direct implication
Fact 3: Within-individual recall variation without new information	Not a primary focus under fixed information sets	Sampling variation averages out under many memory samples	Direct implication
Fact 4: Belief instability without new information	Typically ruled out under dynamic consistency	Not directly implied under canonical formulations	Direct implication
Fact 5: Belief revisions track recall changes	Not directly modeled	Not directly modeled	Direct implication

Memory- and salience-based updating Models of diagnostic expectations (Bordalo, Gennaioli, Ma and Shleifer, 2020b) and related approaches emphasize how salient or representative signals are weighted in forming beliefs. The central object of memory-based frameworks is the recall probability of a stored signal, so stochasticity is not foreign to them; in canonical formulations, however, beliefs are constructed from many samples of the memory database, so that conditional on the database and the cue structure retrieval is effectively deterministic, and the focus has been on the level of beliefs. Our evidence highlights an additional source of variation: stochastic differences

in which stored signals are retrieved at the time beliefs are formed. By abstracting from weighting distortions and isolating retrieval randomness, we clarify how disagreement and belief instability can arise even under common priors and shared information. Stochastic recall is orthogonal to salience-based mechanisms and can operate jointly with them.

Table S.1: EMPIRICAL PATTERNS AND PREDICTIONS OF BELIEF-FORMATION FRAMEWORKS (CONTINUED)

Empirical belief pattern	Informational frictions	Salience / diagnostic models	Stochastic recall
<i>Panel B. Qualitative disagreement experiment (Section 4 of the main paper)</i>			
<i>Disagreement</i>			
Fact 6: Disagreement increases with informational ambivalence	Not a central mechanism	Not a central mechanism	Direct implication
Fact 7: Disagreement decreases with recall effort	Outside canonical formulation	Outside canonical formulation	Direct implication
Fact 8: Asymmetric cue effects on disagreement	Not directly addressed	Partially related through salience amplification	Direct implication
<i>Subjective uncertainty</i>			
Fact 9: Recalled information predicts subjective uncertainty	Not directly addressed	Not directly addressed	Direct implication
Fact 10: Uncertainty increases with informational ambivalence	Not directly addressed	Not directly addressed	Direct implication
Fact 11: Asymmetric cue effects on uncertainty	Not directly addressed	Not directly addressed	Direct implication
<i>Panel C. Qualitative instability experiment (Section 5 of the main paper)</i>			
<i>Belief instability</i>			
Fact 12: Instability increases with informational ambivalence	Not a central mechanism	Not a central mechanism	Direct implication
Fact 13: Instability decreases with recall effort	Outside canonical formulation	Outside canonical formulation	Direct implication
<i>Panel D. Quantitative belief experiment (Section 6 of the main paper)</i>			
<i>Disagreement and instability under dynamic signals</i>			
Fact 14: Disagreement increases with volatility	Not a primary comparative static	Not a primary comparative static	Direct implication
Fact 15: Disagreement increases with persistence	Not a primary comparative static	Not a primary comparative static	Direct implication
Fact 16: Instability increases with volatility	Not directly addressed	Not directly addressed	Direct implication
Fact 17: Instability increases with persistence	Not directly addressed	Not directly addressed	Direct implication

Notes: Entries indicate whether each framework delivers the empirical pattern under its canonical formulation. Descriptions such as “not directly addressed,” “not directly implied under canonical formulations,” or “outside canonical formulation” do not imply inconsistency; extensions of these frameworks could potentially generate some patterns. In particular, the memory-based frameworks contain stochastic retrieval as an ingredient, since their central object is the recall probability of a stored signal; the entries record the focus of canonical formulations rather than what each framework could accommodate. The table highlights that under shared information, the stochastic recall framework delivers all listed patterns as direct implications.

Panels B to D of Table S.1 (continued) provide the analogous comparison for the experimental patterns documented in Sections 4 to 6 of the main paper.

S.2 Stochastic Retrieval versus Noisy Processing: Detailed Comparison

This section expands the comparison in Section 7 of the main paper between our model and models of noisy cognitive processing. Where does the stochasticity in beliefs come from? The cognitive economics literature offers two conceptions. In the first, noise is an exogenous constraint on information processing: the DM receives an imprecise internal impression of the relevant information, a noisy but genuinely informative signal, and then handles it optimally, for instance by shrinkage (Khaw, Li and Woodford, 2021; Woodford, 2020; Azeredo da Silveira, Sung and Woodford, 2024). Beliefs are a deterministic function of the impression, and belief dispersion reflects the intrinsic noise, which processing reduces but cannot eliminate. In the second conception, to which our model belongs, beliefs rest on sampling from memory: the record of stored signals is intact, and stochasticity arises at retrieval, in which signals come to mind when a belief is formed (Bordalo, Gennaioli and Shleifer, 2020a; Bordalo, Conlon, Gennaioli, Kwon and Shleifer, 2023; Enke, Schwelter and Zimmermann, 2024; Conlon and Kwon, 2025). Disagreement, instability, and dispersion in confidence are then microfounded by the sampling process itself. Work in this family has largely studied the first moment of beliefs, in effect averaging over many memory draws, although Bordalo, Conlon, Gennaioli, Kwon and Shleifer (2023) models small-sample recall with the same normal approximation to binomial sampling as Lemma 1 of the main paper and Bordalo, Conlon, Gennaioli, Kwon and Shleifer (2026) study disagreement with a different microfoundation; our contribution is to develop and test the distributional implications of sampling systematically. A third classic source of disagreement, heterogeneous priors, is excluded by design: the experiments induce a common prior.

The two conceptions are separable in our data for two reasons. First, they place the stochasticity at different stages: under processing noise the stored record is corrupted and reading it is deterministic, whereas under stochastic retrieval the record is intact and reading it is stochastic. Second, they differ in what can move beliefs: an internal impression is informative about the history, so beliefs respond only to information and to the noise attached to it, whereas sampling probabilities can also respond to forces that carry no information at all, such as content-free cues. The re-elicitations sharpen the contrast: no information is displayed on Day 2, so imprecision in encoding a present stimulus (Khaw, Li and Woodford, 2021) cannot move beliefs, and beliefs that swing must swing with what comes to mind. Our designs measure recalled content directly and verifiably, yielding four tests, summarized in Table S.2.

1. *Beliefs track measured recall.* Processing noise assigns no role to which specific signals come to mind. In the data, within-person belief revisions co-move with revisions in recalled content nearly one-for-one in the field studies (Result 1 of the main paper) and in the 24-hour re-elicitation (Result 3), and with slopes between one half and one in the dynamic environment (Result 4); the possibility that a common internal impression drives both reports is ruled out by the instrumental-variable and intervention evidence in Section 5.3 of the main paper.

2. *Content-free cues move beliefs through recall.* A cue that carries no information about the state cannot sharpen or tilt a noisy impression of the history, so processing noise predicts no cue effect. In the data, uninformative cues shift the composition of recalled signals and move beliefs and uncertainty in the predicted, asymmetric directions (Result 2 of the main paper).
3. *Forgotten signals return.* Processing noise corrupts the record itself: information absent from the impression is lost and cannot reappear; in the noisy-memory model of Azeredo da Silveira, Sung and Woodford (2024), information not remembered at one date cannot be retrieved at any later date. Under stochastic retrieval, the store is intact, so a signal missed in one draw can surface in the next. In the data, signals absent from a participant’s Day-1 recall re-emerge on Day 2 (Appendix Table A.16 of the main paper); in the quantitative experiment, 90 percent of participants report at least one Day-2 value missing from their Day-1 recall, and most re-emerging values match numbers actually shown (Appendix Table A.21 of the main paper).
4. *Confidence moves with the recalled sample.* Under processing noise, posterior precision is pinned down by the noise parameters and does not vary with the particular realization of the noise. Under stochastic retrieval, confidence depends on the balance of the evidence the DM happens to retrieve. In the data, subjective uncertainty declines with the realized net imbalance of an individual’s recalled evidence (Result 2 of the main paper).

Table S.2: FOUR PATTERNS THAT SEPARATE STOCHASTIC RETRIEVAL FROM NOISY PROCESSING

Pattern	Noisy processing	Stochastic retrieval	Evidence (main paper)
Beliefs track measured recall	No role for which signals come to mind	Beliefs are Bayesian given the realized recall draw	Results 1, 3, and 4; Section 5.3
Content-free cues move beliefs	No channel: an impression reflects information only	Cues shift recall probabilities and hence beliefs	Result 2
Forgotten signals return	Information lost from the impression cannot reappear	Intact store: a signal missed in one draw can surface in the next	Appendix Tables A.16 and A.21
Confidence moves with the recalled sample	Precision fixed by the noise parameters	Uncertainty depends on the balance of recalled evidence	Result 2

Taken together, the evidence supports the memory-based account: belief formation behaves like stochastic sampling from an intact memory store, and this single mechanism generates the observed disagreement, instability, and dispersion in subjective uncertainty. It does so with the primitives the family already has: the central object of memory-based models is a recall probability, and what the evidence adds is that the realized recall draws, and not only their expectation, shape beliefs.

Two qualifications apply. Retrieval effort by itself does not separate the two views, since effort could in principle also reduce processing noise; the separation comes from the item-level signature of the effort treatments, which make recalled counts objectively more accurate while probed item-level recall remains near chance and unchanged by the interventions (Appendix Table A.9 of the main paper), the signature of retrieving more items from an intact store rather than of a uniformly sharper impression. And the two sources of noise are not mutually exclusive: the evidence locates an important part of the stochasticity in beliefs at retrieval without implying that processing noise is absent; Afrouzi, Kwon, Landier, Ma and Thesmar (2023) study an intermediate case in which the stored history is intact but retrieval returns a noisy summary rather than a random subset of its items, and Woodford (2020) notes that for symbolically presented information, noise in storage and retrieval is likely more important than noise in initial perception, exactly the stage our experiments measure and manipulate. Table S.1 in Section S.1 compares informational-friction, salience-based, and stochastic-recall frameworks against the full set of empirical patterns.

S.3 The Mean-Based Approximation for Subjective Uncertainty under Cues

The statements about expected subjective uncertainty in Prediction 3 of the main paper rely on a mean-based approximation, in which $E[U_i]$ is approximated by a decreasing function of $|E[D_i]|$, where $D_i \equiv X_i^+ - X_i^-$. This appendix describes the approximation, explains why it is needed, and reports a numerical verification that the proposition’s comparative statics hold in the relevant experimental parameter ranges.

Exact expression Let $U(D) = \mu(D)(1 - \mu(D))$, where $\mu(D) = [1 + ((1 - b)/b)^D]^{-1}$ is the posterior probability of the good state as a function of recalled net evidence D . Under cueing, X_i^+ and X_i^- are independent binomial random variables with parameters (N^+, p^+) and (N^-, p^-) , where $p^+ = p + a$ if positive signals are cued and p otherwise, and analogously for p^- . The exact expected subjective uncertainty is

$$E[U_i] = \sum_{k=0}^{N^+} \sum_{l=0}^{N^-} \binom{N^+}{k} (p^+)^k (1 - p^+)^{N^+ - k} \binom{N^-}{l} (p^-)^l (1 - p^-)^{N^- - l} U(k - l).$$

This double sum does not admit a closed-form characterization in informational ambivalence A , recall probability p , or cue strength a . Comparative statics of $E[U_i]$ under cueing therefore cannot be derived analytically without further structure.

Mean-based approximation We approximate $E[U_i]$ by $U(E[D_i])$, which is exact when D_i is deterministic and a heuristic approximation otherwise. The approximation is motivated by two observations. First, $U(\cdot)$ is decreasing in $|D|$, so the level of expected uncertainty is determined primarily by the location of the distribution of D_i . Second, the curvature of $U(\cdot)$ is bounded, so the

bias from substituting $U(E[D_i])$ for $E[U_i]$ is bounded by a function of $\text{Var}(D_i)$. The approximation thus delivers tractable comparative statics that capture the directional effect of cueing on expected uncertainty.

Under the mean-based approximation, the threshold for the non-monotonic effect of minority cueing is $\Theta = |N^+ - N^-|p/N^m$, the value of cue strength a at which $E[D_i] = 0$. The uncertainty statements of Prediction 3 of the main paper follow directly: cueing dominant signals shifts $|E[D_i]|$ away from zero (reducing $E[U_i]$); cueing minority signals first shifts $|E[D_i]|$ toward zero (raising $E[U_i]$) and then past zero (reducing $E[U_i]$).

Numerical verification We verify the uncertainty statements of Prediction 3 of the main paper numerically by evaluating the exact $E[U_i]$ at parameter combinations corresponding to the Belief Disagreement Experiment: $N = 20$, $b = 0.65$, $p \in \{0.3, 0.5, 0.7, 0.9\}$, and signal compositions $(N^+, N^-) \in \{(18, 2), (14, 6), (11, 9)\}$ corresponding to informational ambivalence values $A \in \{0.2, 0.6, 0.9\}$. Across all 12 combinations, dominant-signal cueing strictly reduces $E[U_i]$ relative to the no-cue benchmark for every feasible cue strength $a \in (0, 1 - p]$, confirming the dominant-cue statement of Prediction 3. Minority-signal cueing produces a single-peaked profile in a whenever the threshold $\Theta = |N^+ - N^-|p/N^m$ falls inside the feasible range $(0, 1 - p]$, with the exact peak location close to Θ (e.g., $\Theta = 0.40$ vs. exact peak at 0.42 for $(N^+, N^-) = (14, 6)$ and $p = 0.3$; $\Theta = 0.111$ vs. peak at 0.12 for $(N^+, N^-) = (11, 9)$ and $p = 0.5$). When Θ lies outside the feasible range (extreme one-sidedness or high p), $E[U_i]$ is monotonically increasing in a across the entire feasible cue range, consistent with the minority-cue statement of Prediction 3 under the constraint $a < \Theta$.

S.4 Numerical Illustration of Prediction 3 (Effects of Cues on Disagreement)

Figure S.1 illustrates the disagreement statements of Prediction 3 of the main paper for the case when $\lambda < 0.5$. The figure partitions the (a, p) parameter space into regions where cueing positive signals increases, decreases, or has ambiguous effects on cross-sectional disagreement. When $\lambda > 0.5$, any cue that increases the probability of recalling positive signals reduces disagreement.

S.5 Remarks on the Baseline Assumptions

The main analysis focuses on the baseline model with common recall probability, independent recall across stored signals, and Bayesian updating conditional on recalled information. These assumptions provide the cleanest benchmark for isolating the role of imperfect retrieval. The assumption of a constant recall probability can be relaxed in several directions: recall probabilities may depend on cues, recency, rehearsal, primacy, similarity, or interference, and can be endogenized to depend on effort or incentives. Stochastic recall can also be embedded in models that feature other infor-

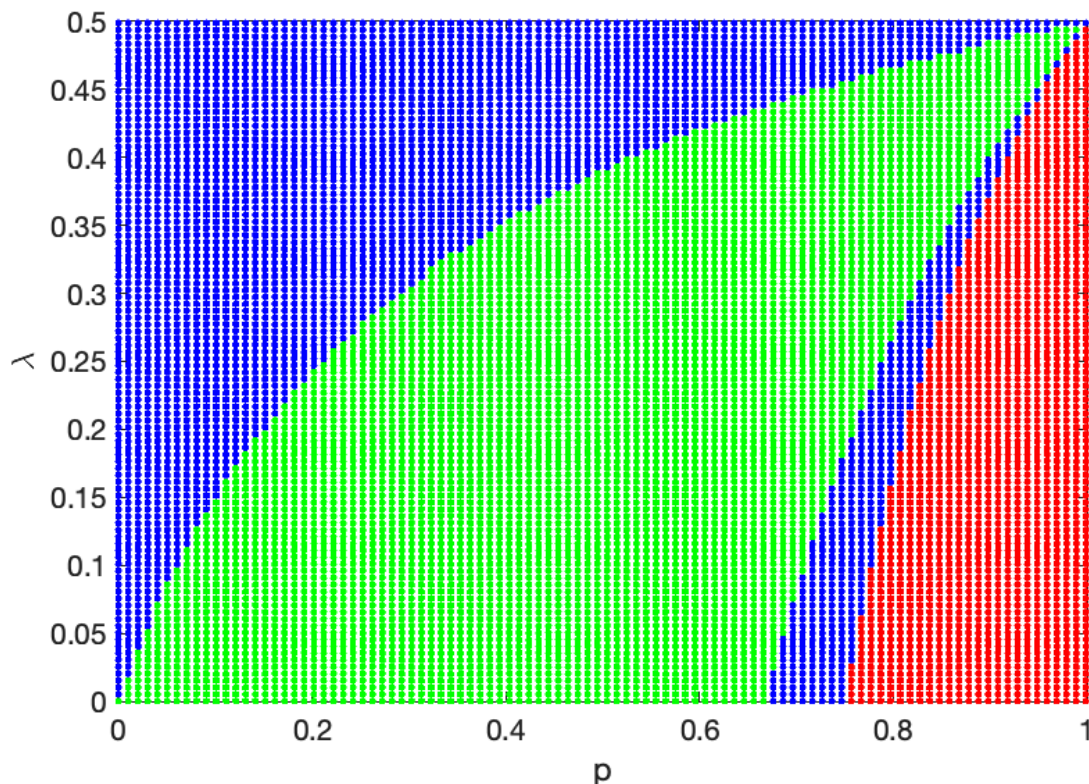


Figure S.1: THE EFFECT OF CUE ON DISAGREEMENT WHEN $\lambda < 0.5$

Notes: In the green area, $\Delta > 1 - p$, so disagreement increases. In the red area, Δ is non-positive, so disagreement decreases. In the blue area, $1 - p > \Delta > 0$, so disagreement increases if $a < \Delta$ and decreases if $a > \Delta$.

mational frictions, such as rational inattention, sticky information, or noisy information, and the Bayesian updating rule can be replaced by alternative rules such as diagnostic expectations. The key distinction in our framework is that the friction enters through retrieval of stored information rather than through signal acquisition or weighting. The extensions below illustrate several of these generalizations.

S.6 Extending the Baseline Model

The baseline framework shows that stochastic recall generates persistent disagreement even when agents observe the same information history. We extend the model along six dimensions that preserve the core stochastic-recall mechanism while broadening its empirical relevance. Specifically, we allow for continuous signals, recency-biased retrieval, probabilistic beliefs, repeated memory sampling, correlated recall across elicitations, and biased priors. Across all extensions, disagreement continues to be shaped by informational ambivalence interacting with imperfect recall, and the model yields sharp predictions about when information interventions reduce dispersion versus unintentionally fragment beliefs.

S.6.1 Continuous Signals

We first generalize the signal structure to continuous realizations, which better captures many macroeconomic and financial information environments. Each decision-maker (DM) observes signals $\{S_k\}_{k=1}^T$ about the binary state $Y \in \{-1, +1\}$, with

$$S_k \mid Y \sim \mathcal{N}(\mu Y, \sigma^2),$$

where $\mu > 0$ measures informativeness.

Under stochastic recall, beliefs depend on the recalled (and potentially time-weighted) signal sum

$$X_i = \sum_{k=1}^T I_k \delta^{T-k} S_k,$$

where $I_k \sim \text{Bernoulli}(p)$ and $\delta \in (0, 1]$ governs temporal weighting.

Informational ambivalence naturally generalizes to

$$\mathcal{A} = 1 - \left| \frac{\sum_{k=1}^T S_k}{\sum_{k=1}^T |S_k|} \right|,$$

which captures how balanced the realized evidence is. As in the baseline model, disagreement increases with ambivalence and decreases with recall accuracy p (S.8).

Associative memory cues operate analogously. Proposition 5 shows that cueing dominant signals unambiguously reduces disagreement, whereas cueing minority signals generates non-monotonic effects: weak cues amplify disagreement by selectively activating different memories across agents, while sufficiently strong cues restore coordination. Thus, modest informational interventions can fragment beliefs when they disproportionately highlight minority evidence.

S.6.2 Recency Effects

To capture recency bias, we allow recalled signals to be temporally weighted by δ^{T-k} . When $\delta < 1$, recent information receives greater weight in belief formation. Proposition 7 shows that recency bias can either increase or decrease disagreement, depending on whether recent signals reinforce or conflict with earlier evidence.

Disagreement is maximized when the recency-weighted evidence is most balanced, so that small differences in which recent observations are recalled translate into large belief dispersion. Proposition 6 further establishes the existence of a disagreement-maximizing recency parameter δ^* , which depends on the realized signal path.

Recency bias also interacts with associative cues: cueing recent minority signals can generate particularly large disagreement by offsetting both numerical and temporal disadvantages in recall.

S.6.3 Probabilistic Beliefs

We next extend belief formation from binary to probabilistic assessments, consistent with survey measures of expectations. Agents report the perceived probability that the state is good as

$$R_i = \frac{X_i^+}{X_i^+ + X_i^-},$$

where X_i^+ and X_i^- denote recalled positive and negative signals (S.7.2).¹

Despite common information, stochastic recall generates dispersion in probabilistic beliefs (Proposition 1). Disagreement continues to rise with informational ambivalence and fall with recall accuracy, mirroring the binary-belief case.

Associative cues again produce asymmetric effects when they target minority signals (Proposition 2). Weak cues increase disagreement by creating heterogeneous recall across agents, whereas sufficiently strong cues reduce dispersion by coordinating retrieval.

S.6.4 Repeated Memory Sampling

The baseline model assumes a single memory draw. We allow agents to sample memory $k \geq 1$ times, which corresponds to recalling from kN^+ positive and kN^- negative signals:

$$X_{i,k}^+ \sim \text{Bin}(kN^+, p), \quad X_{i,k}^- \sim \text{Bin}(kN^-, p).$$

Repeated sampling increases the precision of recalled evidence. S.9 shows that disagreement remains increasing in ambivalence but weakly decreases in k , much as it does with higher recall accuracy. Intuitively, repeated retrieval reduces the role of idiosyncratic sampling variation by increasing overlap in recalled information across agents.

Cue effects are amplified in magnitude but retain the same qualitative structure and thresholds as in the baseline model.

S.6.5 Correlated Recall Across Elicitations

The baseline model draws a fresh, independent recall realization at each elicitation. Retrieval may instead be partly persistent within a person: a signal that came to mind at the first elicitation may be more likely to come to mind again at the second. We capture this with a single parameter $\kappa \in [0, 1]$: at the second elicitation, each stored signal repeats its first-elicitation recall outcome with probability κ and is drawn fresh, with probability p , otherwise. Then κ is exactly the correlation between the two retrieval outcomes of the same signal; $\kappa = 0$ recovers the baseline, and $\kappa = 1$ would freeze the recalled set.

¹The mapping $R_i = X_i^+ / (X_i^+ + X_i^-)$ should be interpreted as a reduced-form rule that translates recalled signal shares into probabilistic beliefs. It coincides (exactly or asymptotically) with the Bayesian posterior mean in a standard Beta–Bernoulli learning problem about the long-run fraction of positive signals. This differs from the baseline binary-state model used in the experiment, but provides a convenient Bayesian rationale for the probabilistic extension.

Correlated recall leaves each elicitation’s recalled evidence with exactly the baseline distribution, so every cross-sectional prediction, in both the binary and the quantitative model, holds verbatim for any κ ; only the joint distribution across the two elicitations changes. For binary beliefs, the switching probability becomes

$$\Pi_\kappa = 2q(1 - q)(1 - \kappa_B),$$

where κ_B is the correlation between the two elicitations’ beliefs induced by κ .² Switching remains strictly positive for every $\kappa < 1$, declines monotonically in κ , and reaches zero only at $\kappa = 1$; we verify numerically that the ambivalence, retrieval-effort, and cue orderings of switching rates across the experimental groups are preserved at every $\kappa \leq 0.95$. In the quantitative model, instability generalizes to $\mathcal{I} \approx 2(1 - \kappa) \frac{1-p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T)$ with $\text{Corr}(\hat{\mu}_{i,1}, \hat{\mu}_{i,2}) \approx \kappa$, so instability stays proportional to the realized variance of the history. In both models the baseline formulas are the $\kappa = 0$ case and an upper bound on instability.

The data place the belief-relevant retrieval draw close to this benchmark. Observed switching rates are near the $\kappa = 0$ prediction $2\hat{q}(1 - \hat{q})$, which under correlated recall is possible only if κ is small at the sign margin of recalled evidence; in the quantitative experiment, revision variances fall modestly short of the independence benchmark, consistent with moderate persistence. At the opposite extreme, $\kappa = 1$ is rejected: signals absent from Day-1 recall frequently re-emerge on Day 2, and recall revisions have positive variance. The strong Day-1-to-Day-2 first stages in the instrumental-variable exercises are too large to reflect correlated retrieval of individual signals given the observed switching; they instead reflect components that are fixed within person, such as encoding, retention, and reporting style (the $e \cdot r \cdot p$ decomposition in the main text). Correlated recall thus reconciles persistent recall reports with fresh retrieval draws: cross-sectional predictions are unaffected for any κ , and belief instability under unchanged information identifies the fresh component of retrieval.

S.6.6 Biased Priors

Finally, we allow agents to hold biased priors $P(G) = \pi \neq \frac{1}{2}$, capturing environments in which individuals begin with asymmetric baseline beliefs. Optimal belief formation then follows a shifted majority rule: agents choose the good state only if net recalled evidence exceeds a prior-dependent threshold $\tau(\pi, b)$.

With biased priors, disagreement is maximized when recalled evidence is balanced relative to this threshold rather than when raw signals are balanced. Moreover, the relationship between recall accuracy and disagreement becomes hump-shaped: disagreement is low when recall is very poor or nearly perfect, and highest at intermediate recall levels.

Associative cues continue to exhibit asymmetric effects. Weak countervailing cues can initially

²The formula follows from $\Pi = \text{Var}(D_1) + \text{Var}(D_2) - 2\text{Cov}(D_1, D_2)$ for the belief indicators D_τ , using that both elicitations share the marginal q . For the quantitative model, the formula in the text follows from $\text{Cov}(I_1^k, I_2^k) = \kappa p(1 - p)$ observation by observation, using the same large- T approximation as the instability proposition in the main text.

increase disagreement by activating minority evidence among some agents, while stronger cues restore coordination by shifting recall more uniformly.

S.7 Properties of the Probabilistic Beliefs

S.7.1 Distribution of the Probabilistic Beliefs

In the baseline model, each DM recalls:

- $X_i^+ \sim \text{Binomial}(N^+, p)$: the number of recalled *positive* signals, and
- $X_i^- \sim \text{Binomial}(N^-, p)$: the number of recalled *negative* signals,

where N^+ and N^- are the total number of positive and negative signals in memory, and p is the recall probability. These two binomial random variables are independent.

We define:

$$R_i = \frac{X_i^+}{X_i^+ + X_i^-}$$

as the proportion of recalled signals that are positive.

Large sample approximation When N^+, N^- are large and p is bounded away from 0 and 1, we may approximate X_i^+ and X_i^- by independent normal random variables:

$$\begin{aligned} X_i^+ &\sim \mathcal{N}(N^+p, N^+p(1-p)), \\ X_i^- &\sim \mathcal{N}(N^-p, N^-p(1-p)). \end{aligned}$$

Applying the delta method to the ratio $h(X_i^+, X_i^-) = X_i^+/(X_i^+ + X_i^-)$ around its mean yields

$$\mathbb{E}[R_i] \approx \frac{N^+}{N^+ + N^-} = \lambda, \quad \text{Var}(R_i) \approx \frac{(1-p)\lambda(1-\lambda)}{Np},$$

where $N = N^+ + N^-$. The variance expression follows from $\text{Var}(R_i) \approx (\partial_+ h)^2 \sigma_+^2 + (\partial_- h)^2 \sigma_-^2$ evaluated at $(\mu_+, \mu_-) = (N^+p, N^-p)$, with $\sigma_\pm^2 = N^\pm p(1-p)$, and simplifies to the displayed form using $N^+ = \lambda N$ and $N^- = (1-\lambda)N$.

S.7.2 Disagreement Based on Probabilistic Beliefs

We extend the analysis to the case with probabilistic beliefs that the asset is good for a certain probability. The probability that the asset is good equals the proportion of recalled positive signals.

Let $R_i = \frac{X_i^+}{X_i^+ + X_i^-}$ denote the proportion of recalled signals that are positive. $X_i^+ \sim \text{Bin}(N^+, p+a)$ and $X_i^- \sim \text{Bin}(N^-, p)$ are independent. The cue increases the probability of recalling each positive signal from p to $p+a$, where $0 < a < 1-p$. Let $\lambda = \frac{N^+}{N^+ + N^-}$ denote the fraction of positive signals, and $N = N^+ + N^-$ be the total number of signals.

The following proposition summarizes the cross-section distribution of probabilistic beliefs and their mean and variance. Moreover, it characterizes the relationship between disagreement and the degree of ambivalence or recall probability.

Proposition 1. (1) (*The cross-sectional distribution of probabilistic beliefs.*) When N is large, R_i is approximately Gaussian around λ , with the variance given in part (2) below.

(2) (*Cross-sectional mean and variance*) By the delta method:

$$\mathbb{E}[R_i] \approx \lambda, \quad \text{Var}(R_i) \approx \frac{(1-p)\lambda(1-\lambda)}{Np}.$$

(3) (*A positive relationship between disagreement and ambivalence.*) The variance of R_i increases in the degree of ambivalence.

(4) (*A negative relationship between disagreement and recall probability.*) The variance of R_i declines with higher recall probability p .

In the probabilistic version of the model, where DMs form beliefs about the asset's value as a probability rather than a binary choice, the cross-sectional distribution of beliefs across DMs can be characterized analytically by the delta method. Specifically, when the total number of signals in memory is large, the proportion of positive signals recalled by a DM—denoted $R_i = X_i^+ / (X_i^+ + X_i^-)$ —is approximately Gaussian around the population proportion λ , with the variance derived from the independent binomial distributions of X_i^+ and X_i^- .

This result allows us to derive expressions for the mean and variance of beliefs across DMs. The mean belief, $\mathbb{E}(R_i)$, converges to the true proportion of positive signals in the memory database, denoted $\lambda = N^+ / N$. However, imperfect memory induces heterogeneity: the variance of beliefs is approximately $\text{Var}(R_i) \approx (1-p)\lambda(1-\lambda)/(Np)$. This expression shows how disagreement depends on both the composition of signals and the probability of recall.

Two key properties emerge from this structure. First, belief disagreement is maximized when the degree of ambivalence in memory is highest—that is, when positive and negative signals are equally balanced ($\lambda = 0.5$). In such cases, small differences in the recalled subset of signals can result in large differences in beliefs across DMs. In contrast, when the memory database is heavily skewed toward either positive or negative signals, even noisy recall tends to produce more uniform beliefs. Second, belief disagreement decreases with the recall probability p . As memory becomes more reliable, DMs are more likely to recall representative samples of the full database, leading to belief convergence. In the limit as $p \rightarrow 1$, all DMs recall all signals and form identical beliefs, eliminating disagreement.

Taken together, these results underscore the central role of memory imperfections in generating heterogeneity in probabilistic beliefs. Even when DMs have access to the same information, variation in recall quality alone can lead to persistent belief dispersion. This mechanism provides a natural explanation for how disagreement arises and persists in settings where objective signals are publicly observed but cognitively filtered through imperfect memory.

S.7.3 The Effects of Cue on Disagreement

The following proposition summarizes the effects of a contextual cue on disagreement in probabilistic beliefs.

Proposition 2 (Cue effects on disagreement in probabilistic beliefs). *The following results summarize the effects of the contextual cue on disagreement in probabilistic beliefs.*

1. If $\lambda \geq \frac{1}{2}$, then the disagreement measured by $\text{Var}(R_i)$ decreases in a . That is, the cue reduces belief dispersion.
2. If $\lambda < \frac{1}{2}$, then the effect of the cue on disagreement is non-monotonic:
 - There exists a unique threshold around $a^* = \frac{p(1-2\lambda)}{\lambda}$ such that:
 - For $a < a^*$, disagreement increases with a : the cue amplifies belief dispersion.
 - For $a > a^*$, disagreement decreases with a : the cue reduces belief dispersion.
 - At $a = a^*$, $\text{Var}(R_i)$ is maximized.
3. If $a^* \geq 1 - p$, then the cue always increases disagreement for all feasible $a \in (0, 1 - p)$. In this case, the cue is too weak (within bounds) to overturn the dominance of negative signals in memory.

We next examine how associative memory influences the cross-sectional disagreement in probabilistic beliefs. In contrast to the binary belief case, where each DM's belief is either positive or negative, we now consider probabilistic beliefs formed by aggregating recalled signals as the proportion of recalled positives: $R_i = \frac{X_i^+}{X_i^+ + X_i^-}$. The disagreement across DMs is measured by the variance of these beliefs, $\text{Var}(R_i)$. We analyze how this disagreement is shaped by contextual cues that enhance recall of positive signals, specifically by increasing the probability of recalling each positive signal from p to $p + a$, where $a > 0$ captures the cue strength.

The analysis reveals that the effect of the cue on disagreement depends crucially on the underlying balance of positive versus negative signals stored in memory, captured by $\lambda = \frac{N^+}{N}$, where N is the total number of signals. If $\lambda \geq \frac{1}{2}$, so that positive signals are at least as common as negative ones in memory, then increasing the cue strength a consistently reduces disagreement. In this case, the cue reinforces the dominant signal type, leading to more consistent recall across agents and hence more homogeneous beliefs.

However, when $\lambda < \frac{1}{2}$, the relationship becomes non-monotonic. For small values of a , the cue causes more variability in which signals are recalled: by increasing the recall of the minority (positive) signal type, the cue pulls some DMs' beliefs away from the dominant (negative) direction. This leads to greater heterogeneity in recalled signal compositions and thus increases disagreement. But as the cue becomes stronger, the recall of positive signals grows more systematic and may begin to offset or even dominate the influence of negative signals. As a result, beliefs begin to reconverge and disagreement falls.

There exists a threshold cue strength a^* at which the variance of probabilistic beliefs is maximized. Below this threshold, the cue amplifies belief heterogeneity; above it, the cue promotes belief convergence. Importantly, the feasibility of this threshold depends on the initial recall probability p : if $a^* \geq 1 - p$, then it lies outside the permissible range of cue strengths, and the cue always increases disagreement within the feasible domain. This case occurs when positive signals are sufficiently scarce in memory, and the cue cannot be made strong enough (without exceeding probability 1) to reverse the dominance of negative recall.

Overall, the results highlight a nuanced role of associative memory cues: they can reduce or exacerbate disagreement depending on the composition of memory and the strength of the cue. This framework provides predictions that can be empirically tested by varying the salience of cues and the composition of signals in memory across experimental treatments.

S.7.4 Proof of Proposition 2

We aim to analyze how the variance of probabilistic beliefs $R_i = \frac{X_i^+}{X_i^+ + X_i^-}$ changes as cue strength a increases, where $X_i^+ \sim \text{Bin}(N^+, p + a)$ and $X_i^- \sim \text{Bin}(N^-, p)$. Let $\lambda = \frac{N^+}{N}$ and $N = N^+ + N^-$.

We approximate the distribution of R_i by applying the delta method. Define:

$$R_i = h(X_i^+, X_i^-) = \frac{X_i^+}{X_i^+ + X_i^-}.$$

Let $\mu_+ = \mathbb{E}[X_i^+] = N^+(p + a) = \lambda N(p + a)$ and $\mu_- = \mathbb{E}[X_i^-] = N^-p = (1 - \lambda)Np$.

Let $\mu = (\mu_+, \mu_-)'$. Then the first-order Taylor expansion of h around μ gives:

$$\mathbb{E}[R_i] \approx h(\mu_+, \mu_-) = \frac{\mu_+}{\mu_+ + \mu_-} = \frac{\lambda(p + a)}{\lambda(p + a) + (1 - \lambda)p}.$$

The variance of R_i is approximated as:

$$\text{Var}(R_i) \approx \nabla h' \Sigma \nabla h,$$

where ∇h is the gradient of h evaluated at μ , and Σ is the covariance matrix of (X_i^+, X_i^-) . Since X_i^+ and X_i^- are independent,

$$\Sigma = \begin{bmatrix} \sigma_+^2 & 0 \\ 0 & \sigma_-^2 \end{bmatrix}, \quad \text{with } \sigma_+^2 = N^+(p + a)(1 - p - a), \quad \sigma_-^2 = N^-p(1 - p).$$

The gradient is:

$$\nabla h = \left(\frac{\mu_-}{(\mu_+ + \mu_-)^2}, \frac{-\mu_+}{(\mu_+ + \mu_-)^2} \right)'.$$

Thus, the variance becomes:

$$\text{Var}(R_i) \approx \left(\frac{\mu_-}{(\mu_+ + \mu_-)^2} \right)^2 \sigma_+^2 + \left(\frac{\mu_+}{(\mu_+ + \mu_-)^2} \right)^2 \sigma_-^2.$$

Substitute $\mu_+ = \lambda N(p + a)$ and $\mu_- = (1 - \lambda)Np$ into this expression. Then:

$$\text{Var}(R_i) \approx \frac{\lambda(1 - \lambda)[(1 - \lambda)p^2(p + a)(1 - p - a) + \lambda(p + a)^2p(1 - p)]}{N(\lambda(p + a) + (1 - \lambda)p)^4}.$$

Let $V(a) = \text{Var}(R_i)$ and compute $\frac{dV}{da}$. We focus on the condition under which the derivative $\frac{dV}{da}$ switches sign. A necessary condition for maximum variance is:

$$\frac{dV}{da} = 0.$$

By simplifying the first-order condition and using symbolic differentiation (detailed in the longer proof), we derive:

$$a^* = \frac{p(1 - 2\lambda)}{\lambda}.$$

- If $\lambda \geq \frac{1}{2}$, then $a^* \leq 0$ is outside the feasible domain ($a > 0$), so $\text{Var}(R_i)$ is decreasing in a .
- If $\lambda < \frac{1}{2}$ and $a^* < 1 - p$, then $\text{Var}(R_i)$ increases in a for $a < a^*$ and decreases for $a > a^*$.
- If $\lambda < \frac{1}{2}$ but $a^* \geq 1 - p$, then for all feasible $a < 1 - p$, $\text{Var}(R_i)$ increases in a .

This completes the proof of Proposition 2.

S.7.5 Evidence from Experiment 1

Experiment 1 of the main paper elicits a probabilistic belief alongside the binary belief, which allows us to examine disagreement based on probabilistic beliefs. Figure S.2 plots two measures by group. Panel (a) shows the cross-sectional variance of the reported probability that the company is good. Panel (b) shows the variance of the binary belief implied by that probability, coded as good when the reported probability is at least 50 percent. The probability-implied measure reproduces all three comparative statics of the main paper: disagreement peaks in the most ambivalent environment, falls with retrieval effort, and moves in opposite directions under the dominant and the minority cue (p values < 0.01). The variance of reported probabilities also rises with informational ambivalence and falls with retrieval effort (p values < 0.05); the magnitudes are smaller because reported probabilities are compressed toward 50 percent, and the cue treatments do not change this measure significantly.

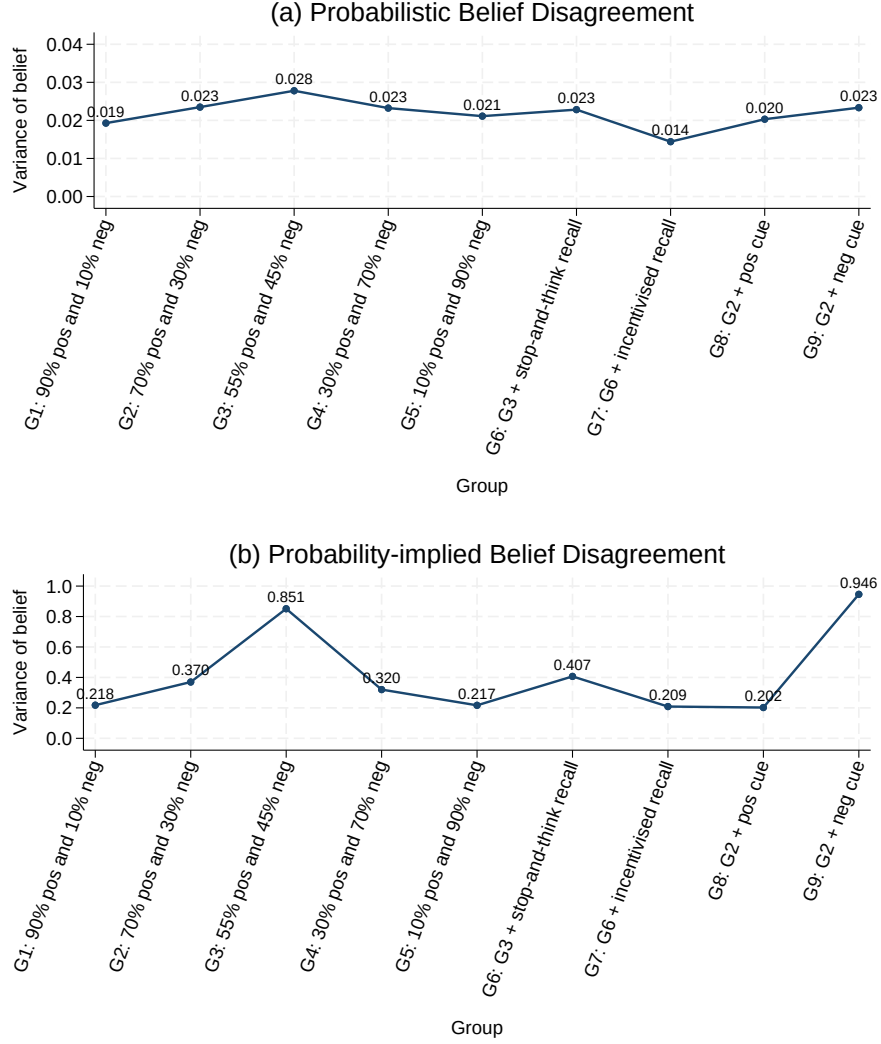


Figure S.2: EXPERIMENT 1: DISAGREEMENT BASED ON PROBABILISTIC BELIEFS

Notes: The figure plots disagreement across the nine groups of Experiment 1 of the main paper. Panel (a): cross-sectional variance of the reported probability $p_i \in [0, 1]$ that the company is good. Panel (b): cross-sectional variance of the probability-implied binary belief, equal to 1 if $p_i \geq 0.5$ and -1 otherwise.

S.7.6 Evidence from Experiment 3

The same check applies to belief instability. Figure S.3 reports two measures based on the probabilistic belief for each group of Experiment 3 of the main paper. Panel (a) shows the cross-sectional variance of within-person revisions in the reported probability between Day 1 and Day 2. It rises with informational ambivalence, from 0.002 in Group 1 to 0.008 in Group 2 and 0.017 in Group 3, and falls to 0.009 with retrieval effort (Group 4); all differences relative to Group 3 are significant (p values < 0.01). Reported probabilities are on average lower on Day 2 than on Day 1 in every group (by 2 to 10 p.p.), a common shift consistent with fewer signals being recalled a day later; instability concerns the dispersion of revisions around this shift. Panel (b) shows the share of

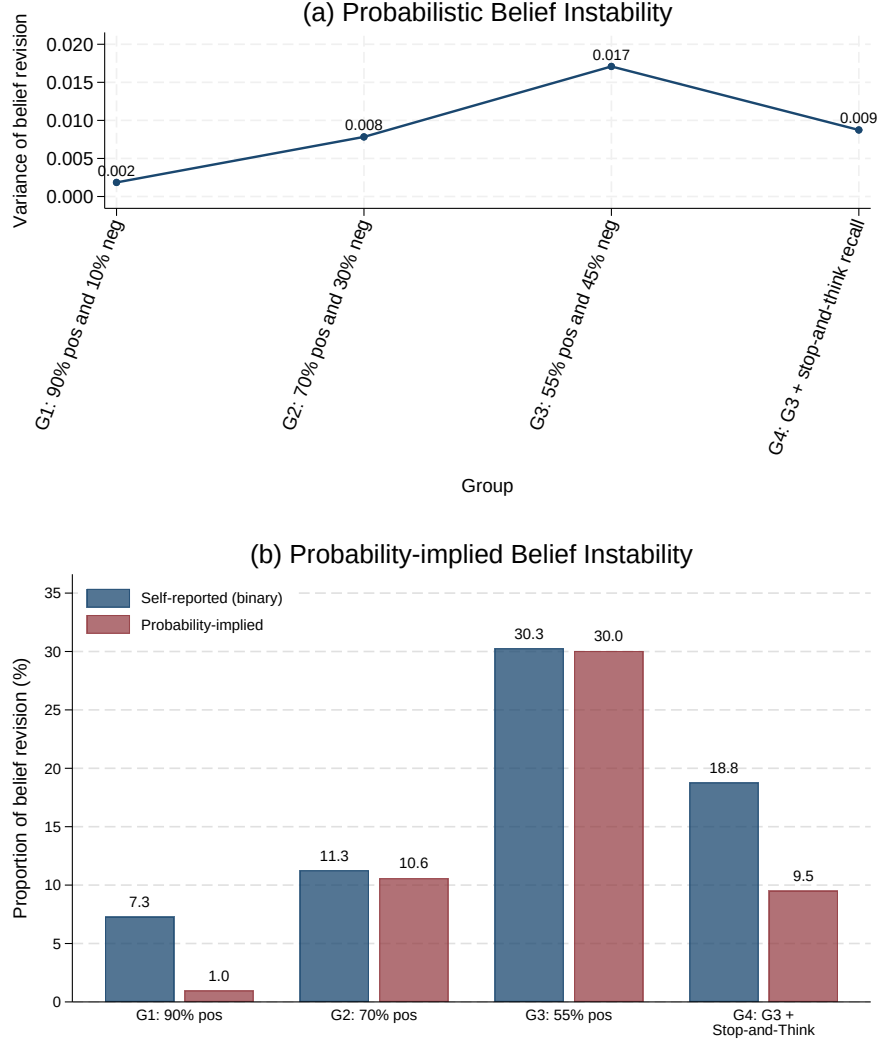


Figure S.3: EXPERIMENT 3: BELIEF INSTABILITY BASED ON PROBABILISTIC BELIEFS

Notes: The figure reports belief instability between Day 1 and Day 2 in Experiment 3 of the main paper, by group. Panel (a): cross-sectional variance of the within-person revision in the reported probability $p_i \in [0, 1]$ that the company is good. Panel (b): proportion of participants whose belief changes; self-reported refers to the reported binary belief, probability-implied to the binary belief implied by the reported probability (good if $p_i \geq 0.5$).

participants whose belief switches between Day 1 and Day 2, using the self-reported binary belief and the binary belief implied by the reported probability (good when the reported probability is at least 50 percent). Probability-implied revision rates are 1.0, 10.6, and 30.0 percent in Groups 1 to 3 and 9.5 percent in Group 4, again with all differences relative to Group 3 significant (p values < 0.01).

S.8 Generalized Model: Continuous Signals and Memory-Based Belief Formation

We generalize the baseline binary signal model to allow for continuous, Gaussian signals and time-decaying recall. This extension retains the core structure of belief formation under imperfect memory, while offering greater analytical tractability and alignment with information-processing frictions observed in practice.

S.8.1 Environment and Belief Formation

Consider a DM who observes a sequence of T diagnostic signals $\{S_k\}_{k=1}^T$ about an unknown binary state $Y \in \{-1, +1\}$. Each signal is drawn conditionally independent from the distribution:

$$S_k \mid Y \sim \mathcal{N}(\mu Y, \sigma^2), \quad \text{for all } k = 1, \dots, T$$

where $\mu > 0$ denotes signal informativeness. The DM encodes each signal into memory upon receipt, but later retrieves only a probabilistic subset due to imperfect and recency-biased recall.

Imperfect recall Each signal S_k is recalled with probability $p \in (0, 1]$, independently across signals.

Recency bias The probability of recall is modulated by a decay factor $\delta \in (0, 1]$ such that older signals are less salient. We implement this by defining a multiplicative recall weight that decays geometrically with signal age:

$$w_k = I_k \cdot \delta^{T-k}, \quad \text{where } I_k \sim \text{Bernoulli}(p)$$

Here, $T - k$ measures how many periods ago signal k arrived. Thus, the DM is more likely to recall recent signals, and their influence is more pronounced even when recalled.

Aggregate evidence The DM's recalled information is aggregated as:

$$Z_i := \sum_{k=1}^T w_k \cdot S_k = \sum_{k=1}^T I_k \cdot \delta^{T-k} \cdot S_k$$

Belief formation The DM forms a binary belief about the asset:

$$\hat{Y}_i = \begin{cases} +1 & \text{if } Z_i > 0 \\ -1 & \text{otherwise} \end{cases}$$

Alternatively, a probabilistic belief $R_i := \Phi(Z_i/\sigma_Z)$ could be used. We focus on binary beliefs here.

S.8.2 Distribution of Beliefs

Under the assumptions above, and for large T , the belief-relevant statistic $Z_i \mid Y$ is approximately normally distributed:

$$Z_i \mid Y \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$$

with:

$$\mu_Z = \mu Y p \sum_{k=1}^T \delta^{T-k} = \mu Y p \cdot \frac{1 - \delta^T}{1 - \delta}, \quad \sigma_Z^2 = \sum_{k=1}^T \delta^{2(T-k)} [p\sigma^2 + p(1-p)(\mu Y)^2] = [p\sigma^2 + p(1-p)\mu^2] \cdot \frac{1 - \delta^{2T}}{1 - \delta^2}$$

Proof. Each term $I_k \delta^{T-k} S_k$ is a product of an independent Bernoulli and a Gaussian random variable. Since I_k and S_k are independent, we apply the law of iterated expectations:

$$\begin{aligned} \mathbb{E}[I_k \delta^{T-k} S_k \mid Y] &= \delta^{T-k} \cdot \mathbb{E}[I_k] \cdot \mathbb{E}[S_k \mid Y] = \delta^{T-k} p \mu Y \\ \text{Var}(I_k \delta^{T-k} S_k \mid Y) &= \mathbb{E} \left[(I_k \delta^{T-k} S_k)^2 \mid Y \right] - \left(\mathbb{E}[I_k \delta^{T-k} S_k \mid Y] \right)^2 \\ &= \delta^{2(T-k)} [\mathbb{E}[I_k] \cdot \mathbb{E}[S_k^2 \mid Y] - p^2 (\mu Y)^2] \\ &= \delta^{2(T-k)} [p(\sigma^2 + (\mu Y)^2) - p^2 (\mu Y)^2] \\ &= \delta^{2(T-k)} [p\sigma^2 + p(1-p)(\mu Y)^2] \end{aligned}$$

Summing over $k = 1$ to T , the total mean and variance are:

$$\begin{aligned} \mu_Z &= \sum_{k=1}^T \delta^{T-k} p \mu Y = \mu Y p \sum_{j=0}^{T-1} \delta^j = \mu Y p \cdot \frac{1 - \delta^T}{1 - \delta} \\ \sigma_Z^2 &= \sum_{k=1}^T \delta^{2(T-k)} [p\sigma^2 + p(1-p)(\mu Y)^2] = [p\sigma^2 + p(1-p)(\mu Y)^2] \cdot \sum_{j=0}^{T-1} \delta^{2j} \\ &= [p\sigma^2 + p(1-p)(\mu Y)^2] \cdot \frac{1 - \delta^{2T}}{1 - \delta^2} \end{aligned}$$

By the Lindeberg-Feller central limit theorem, $Z_i \mid Y$ converges in distribution to a normal variable as T becomes large. $\square$

Proposition 3. (1) (*Distribution of DMs' beliefs*) The belief $\hat{Y}_i$ is distributed as:

$$\hat{Y}_i = \begin{cases} +1 & \text{with probability } q, \\ -1 & \text{with probability } (1 - q) \end{cases} \quad (\text{S.8.1})$$

where

$$q = \Phi \left(\frac{p \sum_{k=1}^T \delta^{T-k} S_k}{\sqrt{p(1-p) \sum_{k=1}^T \delta^{2(T-k)} S_k^2}} \right)$$

(2) (**Cross-sectional mean and variance of DMs' beliefs**) The cross-sectional mean and variance of beliefs are:

$$\mathbb{E}(\hat{Y}_i) = 2q - 1, \quad (\text{S.8.2})$$

$$\text{Var}(\hat{Y}_i) = 4q(1 - q). \quad (\text{S.8.3})$$

Proof. Fix the realized signal sequence $\{S_k\}$ and consider the randomness across DMs arising from independent recall indicators $I_{ik} \sim \text{Bernoulli}(p)$. Define:

$$Z_k = I_{ik} \cdot \delta^{T-k} \cdot S_k, \quad \text{so that} \quad Z_i = \sum_{k=1}^T Z_k.$$

Each Z_k is a random variable that depends only on I_{ik} since S_k is fixed. We derive the moments of Z_k :

Mean:

$$\mathbb{E}[Z_k] = \mathbb{E}[I_{ik}] \cdot \delta^{T-k} S_k = p \cdot \delta^{T-k} S_k$$

Second moment:

$$\mathbb{E}[Z_k^2] = \mathbb{E}[I_{ik}^2] \cdot \left(\delta^{T-k} S_k \right)^2 = p \cdot \delta^{2(T-k)} S_k^2$$

Variance:

$$\begin{aligned} \text{Var}(Z_k) &= \mathbb{E}[Z_k^2] - (\mathbb{E}[Z_k])^2 \\ &= p \cdot \delta^{2(T-k)} S_k^2 - \left(p \cdot \delta^{T-k} S_k \right)^2 \\ &= p(1 - p) \cdot \delta^{2(T-k)} S_k^2. \end{aligned}$$

Then the total statistic $Z_i = \sum_k Z_k$ has:

$$\begin{aligned} \mathbb{E}[Z_i \mid \{S_k\}] &= \sum_{k=1}^T \mathbb{E}[Z_k] = p \sum_{k=1}^T \delta^{T-k} S_k, \\ \text{Var}(Z_i \mid \{S_k\}) &= \sum_{k=1}^T \text{Var}(Z_k) = p(1 - p) \sum_{k=1}^T \delta^{2(T-k)} S_k^2. \end{aligned}$$

Therefore, the probability of a DM choosing +1 is:

$$q = \mathbb{P}(Z_i > 0 \mid \{S_k\}) = \Phi \left(\frac{p \sum_{k=1}^T \delta^{T-k} S_k}{\sqrt{p(1-p) \sum_{k=1}^T \delta^{2(T-k)} S_k^2}} \right).$$

□

S.8.3 Ambivalence, Cues, Recency and Disagreement

Let the sequence of raw signals $\{S_k\}_{k=1}^T$ be fixed and define the *ambivalence index*:

$$\mathcal{A} := 1 - \left| \frac{\sum_{k=1}^T S_k}{\sum_{k=1}^T |S_k|} \right|.$$

This captures the degree of informational balance. When $\sum S_k = 0$, signals are fully ambivalent and $\mathcal{A} = 1$. When all signals point in the same direction, $\mathcal{A} = 0$.

Proposition 4. (*Ambivalence and disagreement*) Holding T , p , δ , and σ^2 fixed, and holding the time pattern of $\{|S_k|\}$ fixed (so that signs are reshuffled without altering when signals of large or small magnitude arrive), the cross-sectional variance of beliefs

$$\text{Var}(\hat{Y}_i) = 4\Phi \left(\frac{\mu_Z}{\sigma_Z} \right) \left[1 - \Phi \left(\frac{\mu_Z}{\sigma_Z} \right) \right]$$

is strictly increasing in the ambivalence index $\mathcal{A}$.

Proof. Define:

$$z := \frac{\mu_Z}{\sigma_Z} = \frac{p \sum_{k=1}^T \delta^{T-k} S_k}{\sqrt{p(1-p) \sum_{k=1}^T \delta^{2(T-k)} S_k^2}}, \quad q := \Phi(z).$$

Let $S_{\text{net}} := \sum_{k=1}^T S_k$ and $S_{\text{mag}} := \sum_{k=1}^T |S_k|$, so the ambivalence index is:

$$\mathcal{A} = 1 - \left| \frac{S_{\text{net}}}{S_{\text{mag}}} \right| \Rightarrow |S_{\text{net}}| = (1 - \mathcal{A}) \cdot S_{\text{mag}}.$$

Assume that the signals are distributed evenly over time (i.e., no time clustering). Then the recency weights δ^{T-k} average out across k , allowing transformation of the weighted sum:

$$\sum_{k=1}^T \delta^{T-k} S_k \approx \bar{w} \cdot S_{\text{net}}, \quad \text{and} \quad \sum_{k=1}^T \delta^{2(T-k)} S_k^2 \approx \hat{w} \cdot \sum_{k=1}^T S_k^2.$$

Substituting gives:

$$|z| \approx \frac{p\bar{w} \cdot |S_{\text{net}}|}{\sqrt{p(1-p) \hat{w} \cdot \sum_{k=1}^T S_k^2}}.$$

The numerator is proportional to $|S_{\text{net}}| = (1 - \mathcal{A}) \cdot S_{\text{mag}}$; the denominator depends only on S_k^2 , which is independent of the sign pattern (and hence of $\mathcal{A}$).

Thus,

$$|z| \propto (1 - \mathcal{A}),$$

which implies $|z|$ is strictly decreasing in $\mathcal{A}$.

Since $\text{Var}(\widehat{Y}_i) = 4\Phi(z)(1 - \Phi(z))$ is strictly decreasing in $|z|$ and maximized at $z = 0$, it follows that:

$$\text{Var}(\widehat{Y}_i) \text{ is strictly increasing in } \mathcal{A}.$$

□

Proposition 5. (*Contextual recall and disagreement.*) Let $w_k := \delta^{T-k}$, and define the recency-weighted net signal

$$E := \sum_{k=1}^T w_k S_k, \quad E^+ := \sum_{k \in \mathcal{P}} w_k S_k,$$

where $\mathcal{P} := \{k : S_k > 0\}$ and $\mathcal{N} := \{k : S_k < 0\}$. Let

$$A := \sum_{k \in \mathcal{P}} w_k^2 S_k^2, \quad B := \sum_{k \in \mathcal{N}} w_k^2 S_k^2,$$

and define

$$r := \frac{E^+}{E}, \quad \alpha := \frac{A}{A + B}.$$

Then:

(a) When $r > 0.5$, a contextual cue that increases recall of positive signals reduces DMs' disagreement about the asset value, i.e.,

$$\text{Var}(\widehat{Y}_i)|_{\text{cue}} < \text{Var}(\widehat{Y}_i)|_{\text{no cue}}.$$

(b) When $r < 0.5$, such a cue may increase or decrease disagreement depending on model parameters. (1) If $\Delta^* > 1 - p$, disagreement increases, i.e.,

$$\text{Var}(\widehat{Y}_i)|_{\text{cue}} > \text{Var}(\widehat{Y}_i)|_{\text{no cue}}.$$

(2) If $1 - p > \Delta^* > 0$, disagreement decreases if $a > \Delta^*$, increases if $a < \Delta^*$, and is unchanged if $a = \Delta^*$. (3) If $\Delta^* \leq 0$, disagreement decreases for any $a > 0$.³

³The disagreement threshold is:

$$\Delta^* = \frac{p(1 - 2p)\alpha - 2p(1 - p)r}{(1 - p)r^2 + p\alpha},$$

where $r = E^+/E$ and $\alpha = A/(A + B)$, with

$$E = \sum_{k=1}^T w_k S_k, \quad E^+ = \sum_{k \in \mathcal{P}} w_k S_k, \quad A = \sum_{k \in \mathcal{P}} w_k^2 S_k^2, \quad B = \sum_{k \in \mathcal{N}} w_k^2 S_k^2,$$

Proof. Let $\{S_k\}_{k=1}^T$ denote a realized sequence of signals, and let $w_k := \delta^{T-k}$ be the recency weight on each signal. Each signal is recalled with a probability that depends on its sign:

$$\mathbb{P}(I_k = 1 \mid S_k > 0) = p + a, \quad \mathbb{P}(I_k = 1 \mid S_k < 0) = p,$$

where $a \in [0, 1 - p]$ captures the cue-induced recall boost for positive signals.

Define the recalled, recency-weighted sum for DM i as

$$Z_i := \sum_{k=1}^T I_k \cdot w_k \cdot S_k,$$

and again let $\mathcal{P} := \{k : S_k > 0\}$, $\mathcal{N} := \{k : S_k < 0\}$. For shorthand, recall the path-level quantities

$$\begin{aligned} E &:= \sum_{k=1}^T w_k S_k, & E^+ &:= \sum_{k \in \mathcal{P}} w_k S_k, \\ A &:= \sum_{k \in \mathcal{P}} w_k^2 S_k^2, & B &:= \sum_{k \in \mathcal{N}} w_k^2 S_k^2. \end{aligned}$$

Mean Under cue and no-cue conditions, the expected signal is:

$$\mathbb{E}[Z_i^{\text{cue}}] = pE + aE^+, \quad \mathbb{E}[Z_i^{\text{no}}] = pE.$$

Variance Since $I_k \in \{0, 1\}$ is independent of S_k , we have:

$$\text{Var}(I_k \cdot w_k S_k) = \mathbb{E}[I_k](1 - \mathbb{E}[I_k])w_k^2 S_k^2.$$

Therefore:

$$\sigma_Z^{2, \text{cue}} = (p + a)(1 - (p + a))A + p(1 - p)B, \quad \sigma_Z^{2, \text{no}} = p(1 - p)(A + B).$$

Disagreement Let the belief index be $z := \mu_Z / \sigma_Z$, and define disagreement as $4\Phi(z)(1 - \Phi(z))$.

We find the threshold Δ^* such that:

$$\frac{\mu_Z^{\text{cue}}}{\sigma_Z^{\text{cue}}} = \frac{\mu_Z^{\text{no}}}{\sigma_Z^{\text{no}}}.$$

The cue leaves disagreement unchanged if the belief index is the same:

$$\frac{\mu_Z^{\text{cue}}}{\sigma_Z^{\text{cue}}} = \frac{\mu_Z^{\text{no}}}{\sigma_Z^{\text{no}}} \iff \frac{pE + aE^+}{\sqrt{(p + a)(1 - (p + a))A + p(1 - p)B}} = \frac{pE}{\sqrt{p(1 - p)(A + B)}}.$$

To eliminate the square roots:

$$\frac{(pE + aE^+)^2}{(p + a)(1 - (p + a))A + p(1 - p)B} = \frac{pE^2}{(1 - p)(A + B)}.$$

$\mathcal{P} = \{k : S_k > 0\}$, $\mathcal{N} = \{k : S_k < 0\}$, and $w_k = \delta^{T-k}$.

$$(pE + aE^+)^2 \cdot (1 - p)(A + B) = pE^2 \cdot [(p + a)(1 - (p + a))A + p(1 - p)B].$$

Expanding both sides and simplifying yields a quadratic in a of the form

$$C_2 a^2 + C_1 a = 0,$$

where C_2 and C_1 are functions of (p, E, E^+, A, B) . The non-trivial root $a = \Delta^*$ leaves disagreement unchanged:

$$\Delta^* = -\frac{C_1}{C_2}.$$

Substituting $r := E^+/E$ and $\alpha := A/(A + B)$ and simplifying gives

$$\Delta^* = \frac{p(1 - 2p)\alpha - 2p(1 - p)r}{(1 - p)r^2 + p\alpha}.$$

Using this threshold, the comparative statics in parts (a) and (b) follow exactly as in the binary, equal-weight case: for $r > 0.5$ (positive signals dominate the recency-weighted evidence), any $a > 0$ moves the belief index further from zero and thus reduces disagreement, while for $r < 0.5$ the sign of the change in disagreement depends on whether a is above or below Δ^* as stated in the proposition. $\square$

Then:

- (a) If $r > 0.5$ (positive signals dominate in the recency-weighted net evidence), then any cue that enhances recall of positive signals ($a > 0$) reduces disagreement:

$$\text{Var}(\hat{Y}_i)|_{\text{cue}} < \text{Var}(\hat{Y}_i)|_{\text{no cue}},$$

as

$$\Phi\left(\frac{pE + aE^+}{\sqrt{(p + a)(1 - (p + a))A + p(1 - p)B}}\right) > \Phi\left(\frac{pE}{\sqrt{p(1 - p)(A + B)}}\right) > 0.5.$$

- (b) If $r < 0.5$ (positive signals are in the minority), the cue's effect on disagreement depends on cue strength:

1. If $\Delta^* > 1 - p$, then disagreement increases for all cue strengths: $\text{Var}(\hat{Y}_i)|_{\text{cue}} > \text{Var}(\hat{Y}_i)|_{\text{no cue}}$.
2. If $0 < \Delta^* < 1 - p$, then disagreement:
 - increases if $a < \Delta^*$,
 - decreases if $a > \Delta^*$,
 - is unchanged if $a = \Delta^*$.
3. If $\Delta^* \leq 0$, then disagreement strictly decreases for all $a > 0$.

Proposition 6 (Disagreement-maximizing recency parameter). *Let $\{S_k\}_{k=1}^T$ denote a fixed sequence of signals and define the belief index under recency decay $\delta \in (0, 1]$ as:*

$$z(\delta) := \sqrt{\frac{p}{1-p}} \frac{E(\delta)}{\sqrt{V(\delta)}} = \frac{p \sum_{k=1}^T \delta^{T-k} S_k}{\sqrt{p(1-p) \sum_{k=1}^T \delta^{2(T-k)} S_k^2}}.$$

Then the cross-sectional variance in binary beliefs:

$$\text{Var}(\hat{Y}_i) = 4\Phi(z(\delta)) \cdot [1 - \Phi(z(\delta))]$$

is maximized at the value $\delta^ \in (0, 1]$ that minimizes the absolute belief index $|z(\delta)|$. In particular:*

- (a) *If there exists a δ^* such that $\sum_{k=1}^T \delta^{T-k} S_k = 0$, then $z(\delta^*) = 0$ and disagreement is maximized.*
- (b) *If no such δ exists, disagreement is maximized at the point(s) where $|z(\delta)|$ is minimized.*

Proof. We aim to characterize the value of δ that maximizes disagreement:

$$\delta^* = \arg \max_{\delta \in (0, 1]} \text{Var}(\hat{Y}_i) = \arg \max_{\delta} [4\Phi(z(\delta)) \cdot (1 - \Phi(z(\delta)))].$$

Define:

$$\text{Var}(\hat{Y}_i)|_{\text{recency}} := 4\Phi(z(\delta))(1 - \Phi(z(\delta))), \quad z(\delta) := \sqrt{\frac{p}{1-p}} \cdot R(\delta), \quad \text{where } R(\delta) := \frac{E(\delta)}{\sqrt{V(\delta)}}.$$

By the chain rule:

$$\frac{dD}{d\delta} = 4 \cdot \phi(z(\delta)) \cdot \frac{dz}{d\delta} \cdot (1 - 2\Phi(z(\delta))).$$

The standard normal density $\phi(z) > 0$ for all z , so the only way $\frac{dD}{d\delta} = 0$ is if:

$$\frac{dz}{d\delta} = 0 \quad \text{or} \quad 1 - 2\Phi(z(\delta)) = 0.$$

The latter implies:

$$\Phi(z(\delta)) = \frac{1}{2} \Rightarrow z(\delta) = 0.$$

This gives the condition that globally maximizes $\text{Var}(\hat{Y}_i)|_{\text{recency}}$:

$$z(\delta^*) = 0 \quad \Longleftrightarrow \quad \sum_{k=1}^T \delta^{T-k} S_k = 0.$$

If such a δ^* exists, it yields the highest possible disagreement. Otherwise, disagreement is strictly below the maximum, and is maximized by minimizing $|z(\delta)|$. $\square$

Proposition 7 (Recency bias and disagreement). *Consider agents who observe T_1 early signals $\{S_{1,j}\}_{j=1}^{T_1}$ and T_2 recent signals $\{S_{2,j}\}_{j=1}^{T_2}$. Define the equal-weighted period aggregates:*

$$\bar{S}_1 = \sum_{j=1}^{T_1} S_{1,j}, \quad \bar{S}_2 = \sum_{j=1}^{T_2} S_{2,j}. \quad (\text{S.8.4})$$

Compare disagreement under two weighting schemes:

- **No recency:** $Z_{eq} = \bar{S}_1 + \bar{S}_2$
- **Recency bias:** $Z_{rec} = \delta^{T_2} \bar{S}_1 + \bar{S}_2$ where $\delta \in (0, 1)$.

The effect of recency bias on disagreement depends on signal alignment and relative aggregate strength:

(a) **Consistent information** ($\bar{S}_1 \bar{S}_2 > 0$): *Recency bias increases disagreement when recent aggregate signals are sufficiently strong ($\delta^{T_2} \bar{S}_1^2 < \bar{S}_2^2$), but decreases disagreement when recent aggregate signals are relatively weak ($\delta^{T_2} \bar{S}_1^2 > \bar{S}_2^2$).*

(b) **Contradictory information** ($\bar{S}_1 \bar{S}_2 < 0$): *Recency bias increases disagreement when recent aggregate signals are relatively weak ($\delta^{T_2} \bar{S}_1^2 > \bar{S}_2^2$), but decreases disagreement when recent aggregate signals are sufficiently strong ($\delta^{T_2} \bar{S}_1^2 < \bar{S}_2^2$).*

The threshold for signal strength becomes $|\bar{S}_2| = \delta^{T_2/2} |\bar{S}_1|$, which depends on both the recency parameter δ and the time gap T_2 between periods.

Proof. Disagreement is measured by $\text{Var}(\hat{Y}_i) = 4\Phi(z)(1 - \Phi(z))$, which is strictly decreasing in $|z|$.

Under no recency:

$$z_{norec} = \frac{p(\bar{S}_1 + \bar{S}_2)}{\sqrt{p(1-p)(\bar{S}_1^2 + \bar{S}_2^2)}}.$$

Under recency bias:

$$z_{rec} = \frac{p(\delta^{T_2} \bar{S}_1 + \bar{S}_2)}{\sqrt{p(1-p)((\delta^{T_2})^2 \bar{S}_1^2 + \bar{S}_2^2)}}.$$

Recency bias increases disagreement when $|z_{rec}| < |z_{eq}|$, which is equivalent to:

$$\frac{|\delta^{T_2} \bar{S}_1 + \bar{S}_2|}{\sqrt{(\delta^{T_2})^2 \bar{S}_1^2 + \bar{S}_2^2}} < \frac{|\bar{S}_1 + \bar{S}_2|}{\sqrt{\bar{S}_1^2 + \bar{S}_2^2}}.$$

Squaring both sides and cross-multiplying:

$$(\delta^{T_2} \bar{S}_1 + \bar{S}_2)^2 (\bar{S}_1^2 + \bar{S}_2^2) < (\bar{S}_1 + \bar{S}_2)^2 ((\delta^{T_2})^2 \bar{S}_1^2 + \bar{S}_2^2).$$

Following the algebraic expansion and simplification from the two-period case, this reduces to:

$$2\delta^{T_2} \bar{S}_1 \bar{S}_2 (\bar{S}_1^2 + \bar{S}_2^2) < 2\bar{S}_1 \bar{S}_2 ((\delta^{T_2})^2 \bar{S}_1^2 + \bar{S}_2^2).$$

Case 1: Consistent aggregates ($\bar{S}_1 \bar{S}_2 > 0$).

Dividing both sides by $2\bar{S}_1\bar{S}_2 > 0$ (inequality direction unchanged):

$$\delta^{T_2}(\bar{S}_1^2 + \bar{S}_2^2) < (\delta^{T_2})^2\bar{S}_1^2 + \bar{S}_2^2.$$

Expanding and rearranging:

$$\begin{aligned}\delta^{T_2}\bar{S}_1^2 + \delta^{T_2}\bar{S}_2^2 &< (\delta^{T_2})^2\bar{S}_1^2 + \bar{S}_2^2 \\ \delta^{T_2}\bar{S}_1^2 - (\delta^{T_2})^2\bar{S}_1^2 &< \bar{S}_2^2 - \delta^{T_2}\bar{S}_2^2 \\ \delta^{T_2}\bar{S}_1^2(1 - \delta^{T_2}) &< \bar{S}_2^2(1 - \delta^{T_2}) \\ (1 - \delta^{T_2})(\delta^{T_2}\bar{S}_1^2 - \bar{S}_2^2) &< 0.\end{aligned}$$

Since $(1 - \delta^{T_2}) > 0$ for $\delta \in (0, 1)$, we need:

$$\delta^{T_2}\bar{S}_1^2 - \bar{S}_2^2 < 0 \quad \Rightarrow \quad \delta^{T_2}\bar{S}_1^2 < \bar{S}_2^2.$$

Case 2: Contradictory aggregates ($\bar{S}_1\bar{S}_2 < 0$).

Dividing both sides by $2\bar{S}_1\bar{S}_2 < 0$ (inequality direction flips):

$$\delta^{T_2}(\bar{S}_1^2 + \bar{S}_2^2) > (\delta^{T_2})^2\bar{S}_1^2 + \bar{S}_2^2.$$

Following the same algebraic steps:

$$(1 - \delta^{T_2})(\delta^{T_2}\bar{S}_1^2 - \bar{S}_2^2) > 0.$$

Since $(1 - \delta^{T_2}) > 0$, we need:

$$\delta^{T_2}\bar{S}_1^2 - \bar{S}_2^2 > 0 \quad \Rightarrow \quad \delta^{T_2}\bar{S}_1^2 > \bar{S}_2^2.$$

Therefore, recency bias increases disagreement when recent aggregate signals are either sufficiently strong in consistent environments or sufficiently weak in contradictory environments, both conditions leading to reduced informativeness of the weighted signal aggregate. $\square$

The conditions reveal that recency bias affects disagreement through different mechanisms depending on signal alignment across periods. When period aggregates are aligned (same signs), recency bias increases disagreement when recent aggregate signals are sufficiently strong ($|\bar{S}_2| > \delta^{T_2/2}|\bar{S}_1|$). In this case, overweighting strong recent period information while heavily discounting historical period information reduces total signal strength. When period aggregates are contradictory (opposite signs), recency bias increases disagreement when recent aggregate signals are sufficiently weak ($|\bar{S}_2| < \delta^{T_2/2}|\bar{S}_1|$). Here, underweighting strong contradictory historical period information while maintaining weak recent period signals leads to less informative belief formation. In both cases, recency bias increases disagreement when it results in a less informative weighted signal

aggregate, moving beliefs closer to the uninformative prior and thereby increasing cross-sectional dispersion.

The threshold $\delta^{T_2/2}$ captures the interaction between the recency parameter, signal strength, and temporal structure. As recency bias intensifies (δ decreases) or the time gap between periods increases (larger T_2), the threshold for recent signals to trigger increased disagreement becomes lower, making disagreement more likely in aligned signal environments and less likely in contradictory signal environments. This highlights how both the degree of recency bias and the temporal spacing of information arrival jointly determine the welfare effects of memory-based information processing.

In this appendix we extend the analysis to the case of biased priors and derive the analogue of Propositions 2, 3, and 4 under the *Bayes-optimal* belief formation rule.

S.8.4 Setup with Biased Priors

The state is binary $Y \in \{G, B\}$ with prior

$$P(G) = \pi, \quad P(B) = 1 - \pi, \quad \pi \in (0, 1).$$

Signals $s_k \in \{p, n\}$ (positive/negative) are i.i.d. and symmetrically diagnostic:

$$P(p \mid G) = P(n \mid B) = b > \frac{1}{2}, \quad P(p \mid B) = P(n \mid G) = 1 - b.$$

The DM observes N^+ positive and N^- negative signals in the learning phase. In the recall phase, each signal is independently recalled with probability p . Let

$$X_i^+ \sim \text{Bin}(N^+, p), \quad X_i^- \sim \text{Bin}(N^-, p),$$

denote the numbers of recalled positive and negative signals for agent i . Define the difference

$$D_i := X_i^+ - X_i^-.$$

The agent chooses $a_i \in \{G, B\}$ and incurs 0–1 loss

$$L(a_i, Y) = \mathbf{1}\{a_i \neq Y\}.$$

S.8.5 Optimal Belief Formation Rule

Given recalled counts (X_i^+, X_i^-) , the likelihoods are

$$P(R_i \mid G) = b^{X_i^+} (1 - b)^{X_i^-}, \quad P(R_i \mid B) = (1 - b)^{X_i^+} b^{X_i^-}.$$

Posterior odds are

$$\frac{P(G | R_i)}{P(B | R_i)} = \frac{P(G)}{P(B)} \cdot \frac{P(R_i | G)}{P(R_i | B)} = \frac{\pi}{1 - \pi} \left(\frac{b}{1 - b} \right)^{X_i^+ - X_i^-}.$$

Under symmetric 0–1 loss, the Bayes–optimal decision is

$$\text{choose } G \iff P(G | R_i) > \frac{1}{2} \iff \frac{P(G | R_i)}{P(B | R_i)} > 1.$$

Since $\log\left(\frac{b}{1-b}\right) > 0$, this is equivalent to

$$\log \frac{\pi}{1 - \pi} + (X_i^+ - X_i^-) \log \frac{b}{1 - b} > 0.$$

Define the prior–signal threshold

$$\tau(\pi, b) := \frac{\log \frac{1 - \pi}{\pi}}{\log \frac{b}{1 - b}}. \quad (\text{S.8.5})$$

Then the optimal rule can be written as

$$a_i^* = \begin{cases} G & \text{if } D_i > \tau(\pi, b), \\ B & \text{otherwise.} \end{cases} \quad (\text{S.8.6})$$

Equivalently, the agent behaves as if applying a *shifted majority rule* over recalled signals, where the shift depends on the prior π and signal accuracy b .

For notational convenience we encode the binary belief as

$$Y_i^{b, \pi} := \begin{cases} 1 & \text{if } D_i > \tau(\pi, b), \\ -1 & \text{otherwise.} \end{cases}$$

S.8.6 Distribution of Beliefs

For large N^+, N^- , we approximate the binomial recall counts by normals:

$$X_i^+ \approx \mathcal{N}(N^+ p, N^+ p(1 - p)), \quad X_i^- \approx \mathcal{N}(N^- p, N^- p(1 - p)),$$

independently. Hence

$$D_i = X_i^+ - X_i^- \approx \mathcal{N}(\mu_D, \sigma_D^2),$$

with

$$\mu_D = (N^+ - N^-)p, \quad \sigma_D^2 = p(1 - p)N, \quad N := N^+ + N^-.$$

Define

$$z := \frac{\mu_D - \tau(\pi, b)}{\sigma_D}.$$

Let

$$q_\pi := P(Y_i^{b,\pi} = 1) = P(D_i > \tau(\pi, b)) = \Phi(z),$$

where Φ is the standard normal cdf. Then

$$Y_i^{b,\pi} = \begin{cases} 1 & \text{w.p. } q_\pi, \\ -1 & \text{w.p. } 1 - q_\pi, \end{cases} \Rightarrow \text{Var}(Y_i^{b,\pi}) = 4q_\pi(1 - q_\pi) = 4\Phi(z)(1 - \Phi(z)).$$

Note that $\text{Var}(Y_i^{b,\pi})$ is a single-peaked function of z , with a unique maximum at $z = 0$.

S.8.7 Biased-Prior Analogue of Prediction 1

In the baseline (unbiased) model, disagreement is increasing in the “ambivalence” of the information, defined by how balanced N^+ and N^- are. With biased priors, what matters is how close the *expected net recalled evidence* μ_D is to the cutoff $\tau(\pi, b)$ implied by the prior.

We capture this with an *effective ambivalence* index:

$$A^\pi := 1 - \frac{|(N^+ - N^-)p - \tau(\pi, b)|}{pN}.$$

Note that $A^\pi \in [0, 1]$ and A^π is maximal when $(N^+ - N^-)p = \tau(\pi, b)$.

Proposition 8 (Biased-Prior Ambivalence and Disagreement). *Fix N , recall probability p , prior π , and signal accuracy b . Then, under the optimal rule (S.8.6),*

$$\text{Var}(Y_i^{b,\pi}) = 4\Phi(z)(1 - \Phi(z)), \quad z = \frac{(N^+ - N^-)p - \tau(\pi, b)}{\sqrt{p(1-p)N}},$$

is strictly decreasing in $|(N^+ - N^-)p - \tau(\pi, b)|$, and hence strictly increasing in A^π . Disagreement is maximized when

$$(N^+ - N^-)p = \tau(\pi, b).$$

Proof (sketch). For fixed N, p, π, b , both $\tau(\pi, b)$ and σ_D are constants, so

$$z = c_1((N^+ - N^-)p - \tau(\pi, b)),$$

for some positive constant c_1 . Thus $|z|$ is strictly increasing in $|(N^+ - N^-)p - \tau(\pi, b)|$.

The function $f(z) := 4\Phi(z)(1 - \Phi(z))$ is symmetric in z and strictly decreasing in $|z|$, with unique maximum at $z = 0$. Therefore $\text{Var}(Y_i^{b,\pi}) = f(z)$ is strictly decreasing in $|(N^+ - N^-)p - \tau(\pi, b)|$ and strictly increasing in A^π . It is maximized when $z = 0$, i.e. when $(N^+ - N^-)p = \tau(\pi, b)$. $\square$

When $\pi = \frac{1}{2}$, we have $\tau(\pi, b) = 0$ and A^π reduces to the baseline ambivalence measure, so Proposition 8 collapses to the original Prediction 1 of the main paper.

S.8.8 Biased-Prior Analogue of Prediction 2

In the unbiased model, disagreement is decreasing in the recall probability p . With biased priors, disagreement vanishes when recall is extremely poor ($p \rightarrow 0$) or nearly perfect ($p \rightarrow 1$), and exhibits a hump shape in between.

Proposition 9 (Disagreement and Recall with Biased Priors). *Fix N^+, N^-, π, b . Under the optimal biased-prior rule (S.8.6), disagreement $\text{Var}(Y_i^{b,\pi})$ as a function of $p \in (0, 1)$ satisfies:*

- (i) $\text{Var}(Y_i^{b,\pi}) \rightarrow 0$ as $p \rightarrow 1$.
- (ii) $\text{Var}(Y_i^{b,\pi}) \rightarrow 0$ as $p \rightarrow 0$.
- (iii) *There exists at least one $p^* \in (0, 1)$ such that $\text{Var}(Y_i^{b,\pi})$ is strictly positive for p in a neighborhood of p^* and attains a maximum there.*

Moreover, in the normal approximation, p^* is characterized by the condition

$$(N^+ - N^-)p^* \approx \tau(\pi, b).$$

Proof (sketch). Recall that

$$z(p) = \frac{(N^+ - N^-)p - \tau(\pi, b)}{\sqrt{p(1-p)N}}, \quad \text{Var}(Y_i^{b,\pi}) = 4\Phi(z(p))(1 - \Phi(z(p))).$$

(i) As $p \rightarrow 1$, $\sigma_D^2 = p(1-p)N \rightarrow 0$, so D_i becomes essentially deterministic with limit $D_i \rightarrow N^+ - N^-$. Hence $z(p) \rightarrow \pm\infty$ depending on the sign of $(N^+ - N^-) - \tau(\pi, b)$. Thus $q_\pi = \Phi(z(p)) \rightarrow 0$ or 1, and $\text{Var}(Y_i^{b,\pi}) \rightarrow 0$.

(ii) As $p \rightarrow 0$, we have $\mu_D = (N^+ - N^-)p \rightarrow 0$ and $\sigma_D^2 = p(1-p)N \rightarrow 0$, while $\tau(\pi, b)$ is fixed. Therefore

$$z(p) = \frac{\mu_D - \tau(\pi, b)}{\sigma_D} \sim -\frac{\tau(\pi, b)}{\sqrt{pN}} \rightarrow \begin{cases} +\infty & \text{if } \tau(\pi, b) < 0, \\ -\infty & \text{if } \tau(\pi, b) > 0. \end{cases}$$

In either case, $|z(p)| \rightarrow \infty$, so $q_\pi \rightarrow 0$ or 1 and $\text{Var}(Y_i^{b,\pi}) \rightarrow 0$. Intuitively, with very low recall, everyone falls back on the prior.

(iii) By continuity of $z(p)$ on $(0, 1)$ and the intermediate value theorem, there exists some $p^* \in (0, 1)$ such that $z(p^*) = 0$, i.e.

$$(N^+ - N^-)p^* = \tau(\pi, b).$$

At such p^* , we have $q_\pi = \frac{1}{2}$ and hence $\text{Var}(Y_i^{b,\pi}) = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$, the maximal possible value. Since $\text{Var}(Y_i^{b,\pi}) \rightarrow 0$ at both endpoints $p \rightarrow 0$ and $p \rightarrow 1$ and is continuous in p , it must attain a local maximum at some $p^* \in (0, 1)$.

In the normal approximation, this p^* is characterized by $z(p^*) = 0$, giving $(N^+ - N^-)p^* = \tau(\pi, b)$ as claimed. $\square$

In contrast to the unbiased case ($\tau = 0$), disagreement is no longer monotonically decreasing in p . Instead it is hump-shaped: it is low when recall is extremely poor (agents rely almost purely on the prior) and when recall is nearly perfect (agents see almost the same evidence), and highest at intermediate recall probabilities where recalled signals and priors just offset.

S.8.9 Biased-Prior Analogue of Prediction 3 (Contextual Cues)

We now consider the effect of an associative cue that selectively increases recall of one signal type. For concreteness, suppose a cue increases recall of positive signals from p to $p+a$, with $a \in [0, 1-p]$.

Under the cue:

$$X_i^{+, \text{cue}} \approx \mathcal{N}(N^+(p+a), N^+(p+a)(1-p-a)), \quad X_i^{-, \text{cue}} \approx \mathcal{N}(N^-p, N^-p(1-p)),$$

independently. Hence

$$D_i^{\text{cue}} := X_i^{+, \text{cue}} - X_i^{-, \text{cue}} \approx \mathcal{N}(\mu_D^{\text{cue}}, (\sigma_D^{\text{cue}})^2),$$

with

$$\mu_D^{\text{cue}} = N^+(p+a) - N^-p, \quad (\sigma_D^{\text{cue}})^2 = N^+(p+a)(1-p-a) + N^-p(1-p).$$

Define

$$z^{\text{cue}} := \frac{\mu_D^{\text{cue}} - \tau(\pi, b)}{\sigma_D^{\text{cue}}}, \quad q_\pi^{\text{cue}} := P(D_i^{\text{cue}} > \tau(\pi, b)) = \Phi(z^{\text{cue}}).$$

Then

$$\text{Var}(Y_i^{b, \pi} \mid \text{cue}) = 4q_\pi^{\text{cue}}(1 - q_\pi^{\text{cue}}) = 4\Phi(z^{\text{cue}})(1 - \Phi(z^{\text{cue}})).$$

Recall that $\text{Var}(Y)$ is strictly decreasing in $|z|$; thus the sign of the effect of the cue on disagreement is governed by how $|z|$ changes when we move from (μ_D, σ_D) to $(\mu_D^{\text{cue}}, \sigma_D^{\text{cue}})$.

Proposition 10 (Associative Cues with Biased Priors). *Fix N^+, N^-, p, π, b and let $\tau = \tau(\pi, b)$ be defined by (S.8.5). Consider a cue that increases recall of positive signals from p to $p+a$.*

- (i) *If the cue moves the expected difference μ_D further away from the threshold τ in standardized units, i.e.*

$$|z^{\text{cue}}| > |z|,$$

then disagreement strictly decreases:

$$\text{Var}(Y_i^{b, \pi} \mid \text{cue}) < \text{Var}(Y_i^{b, \pi} \mid \text{no cue}).$$

This occurs when the cue strengthens a “dominant direction” (signals whose majority is already aligned with the prior/posterior cut).

- (ii) *If the cue initially moves μ_D closer to τ in standardized units, i.e.*

$$|z^{\text{cue}}| < |z| \quad \text{for small } a > 0,$$

then for sufficiently small a the cue increases disagreement, while for sufficiently large a it eventually decreases disagreement. There exists a threshold cue strength $\Delta_\pi \in (0, 1 - p)$ such that $\text{Var}(Y_i^{b,\pi})$ is maximized at $a = \Delta_\pi$.

Proof (sketch). The variance of beliefs is

$$\text{Var}(Y_i^{b,\pi}) = f(z), \quad f(z) := 4\Phi(z)(1 - \Phi(z)),$$

and f is strictly decreasing in $|z|$. Under the cue, z is replaced by z^{cue} .

(i) If $|z^{\text{cue}}| > |z|$, then $f(z^{\text{cue}}) < f(z)$ by monotonicity of f in $|z|$, so disagreement decreases.

(ii) Suppose for small $a > 0$ we have $|z^{\text{cue}}| < |z|$. Then $f(z^{\text{cue}}) > f(z)$, so disagreement increases initially. As a grows, the mean μ_D^{cue} eventually moves far to the right of τ , while the change in $(\sigma_D^{\text{cue}})^2$ is bounded, so $|z^{\text{cue}}| \rightarrow \infty$ and $f(z^{\text{cue}}) \rightarrow 0$. Thus disagreement must eventually decrease for large a . Continuity of $f(z^{\text{cue}}(a))$ in a implies that there exists at least one $\Delta_\pi \in (0, 1 - p)$ at which $f(z^{\text{cue}}(a))$ attains a local maximum, i.e. disagreement is maximized. $\square$

When $\pi = \frac{1}{2}$ (so $\tau = 0$), Proposition 10 reduces to the baseline Prediction 3 of the main paper: cueing a signal type that is already dominant strictly reduces disagreement, while cueing a minority signal type yields a non-monotone effect with a threshold cue strength separating disagreement-increasing from disagreement-decreasing regimes.

S.9 Sampling Memory Multiple Times

This appendix extends the baseline recall process by allowing each DM to sample her memory database $k \geq 1$ independent times. Each sampling round recalls each past signal with probability p , independently across rounds and signals. Equivalently, each original signal generates k independent recall opportunities, so that the recalled counts are distributed as if the DM had observed kN^+ positive and kN^- negative signals, each recalled with probability p .

Let N^+ and N^- denote the numbers of positive and negative signals generated in the learning phase, and let $N := N^+ + N^-$. Define $X_{i,k}^+$ and $X_{i,k}^-$ as the numbers of positive and negative signals recalled by DM i after k sampling rounds. The DM applies the same majority rule as in the baseline: $\hat{Y}_i = 1$ iff $X_{i,k}^+ > X_{i,k}^-$, and $\hat{Y}_i = -1$ otherwise.

Proposition 11 (k -sampling analogue of Lemma 1 of the main paper). *Assume N^+ and N^- are large. Under k -fold memory sampling,*

$$X_{i,k}^+ \sim \mathcal{N}(kN^+p, kN^+p(1-p)),$$

$$X_{i,k}^- \sim \mathcal{N}(kN^-p, kN^-p(1-p)),$$

independently. Let $Z_{i,k} := X_{i,k}^+ - X_{i,k}^-$. Then

$$Z_{i,k} \sim \mathcal{N}(k(N^+ - N^-)p, kp(1-p)N).$$

Hence the probability that DM i forms the good-state belief is

$$q_k \equiv P(X_{i,k}^+ > X_{i,k}^-) = \Phi\left(\sqrt{k} \frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}\right).$$

The cross-sectional mean and variance of beliefs are

$$E(\widehat{Y}_i) = 2q_k - 1, \quad \text{Var}(\widehat{Y}_i) = 4q_k(1 - q_k).$$

Proof. Each original positive (negative) signal is recalled independently in each of k rounds with probability p . Therefore,

$$X_{i,k}^+ \sim \text{Bin}(kN^+, p), \quad X_{i,k}^- \sim \text{Bin}(kN^-, p),$$

independently. For large N^+ and N^- , the binomials admit normal approximations:

$$\begin{aligned} X_{i,k}^+ &\approx \mathcal{N}(kN^+p, kN^+p(1-p)), \\ X_{i,k}^- &\approx \mathcal{N}(kN^-p, kN^-p(1-p)). \end{aligned}$$

Define $Z_{i,k} = X_{i,k}^+ - X_{i,k}^-$. The difference of independent normals is normal, so

$$Z_{i,k} \sim \mathcal{N}(\mu_{Z,k}, \sigma_{Z,k}^2),$$

with

$$\mu_{Z,k} = k(N^+ - N^-)p, \quad \sigma_{Z,k}^2 = kN^+p(1-p) + kN^-p(1-p) = kp(1-p)N.$$

Thus,

$$\begin{aligned} q_k &= P(X_{i,k}^+ > X_{i,k}^-) = P(Z_{i,k} > 0) = \Phi\left(\frac{\mu_{Z,k}}{\sigma_{Z,k}}\right) \\ &= \Phi\left(\frac{k(N^+ - N^-)p}{\sqrt{k} \sqrt{p(1-p)N}}\right) = \Phi\left(\sqrt{k} \frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}\right). \end{aligned}$$

Since $\widehat{Y}_i = 1$ with probability q_k and -1 otherwise,

$$E(\widehat{Y}_i) = 1 \cdot q_k + (-1) \cdot (1 - q_k) = 2q_k - 1,$$

and because $\widehat{Y}_i^2 = 1$ always,

$$\text{Var}(\widehat{Y}_i) = E(\widehat{Y}_i^2) - E(\widehat{Y}_i)^2 = 1 - (2q_k - 1)^2 = 4q_k(1 - q_k).$$

□

Proposition 12 (k -sampling analogue of Propositions 2 and 3). *Fix $k \geq 1$ and $p \in (0, 1)$.*

(i) (Ambivalence.) Disagreement $\text{Var}(\hat{Y}_i)$ is strictly increasing in the degree of ambivalence

$$A := 1 - \left| \frac{N^+ - N^-}{N} \right|.$$

(ii) (Memory effort / sampling intensity.) For fixed (N^+, N^-, p) , disagreement is weakly decreasing in k , with strict decrease whenever $N^+ \neq N^-$. In particular, for any $k > 1$,

$$\text{Var}(\hat{Y}_i; k) \leq \text{Var}(\hat{Y}_i; 1).$$

Proof. From Proposition 11, let

$$c_k := \sqrt{k} \frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}, \quad q_k = \Phi(c_k), \quad \text{Var}(\hat{Y}_i) = 4q_k(1 - q_k) =: f(c_k).$$

The function $f(z) = 4\Phi(z)(1 - \Phi(z))$ is symmetric and strictly decreasing in $|z|$, with a unique maximum at $z = 0$.

Part (i). For fixed (p, k, N) , $|c_k|$ is proportional to $\left| \frac{N^+ - N^-}{N} \right|$. Since f decreases in $|c_k|$, disagreement increases when $\left| \frac{N^+ - N^-}{N} \right|$ falls. Equivalently, disagreement is strictly increasing in

$$A = 1 - \left| \frac{N^+ - N^-}{N} \right|.$$

Part (ii). For fixed (N^+, N^-, p) ,

$$|c_k| = \sqrt{k} |c_1|.$$

If $N^+ \neq N^-$ then $|c_1| > 0$, so $|c_k| > |c_1|$ for all $k > 1$. Since f strictly decreases in $|z|$, we have

$$f(c_k) < f(c_1) \quad \Rightarrow \quad \text{Var}(\hat{Y}_i; k) < \text{Var}(\hat{Y}_i; 1).$$

If $N^+ = N^-$ then $c_k = 0$ for all k and the variances coincide. This proves the claim. $\square$

Proposition 13 (k -sampling analogue of Prediction 3 of the main paper (Contextual Cues)). Suppose a contextual cue is associated with positive signals and increases their recall probability from p to $\tilde{p} = p + a$, where $a \in [0, 1 - p]$. Negative signals remain recalled with probability p . Under k -fold sampling,

$$\tilde{q}_k \equiv P(\hat{Y}_i = 1 \mid \text{cue}) = \Phi \left(\sqrt{k} \frac{N^+ \tilde{p} - N^- p}{\sqrt{N^+ \tilde{p}(1 - \tilde{p}) + N^- p(1 - p)}} \right).$$

Let q_k be as in Proposition 11. Then:

(i) If $N^+ \geq N^-$, then $\tilde{q}_k > q_k > \frac{1}{2}$, so the cue strictly reduces disagreement:

$$\text{Var}(\hat{Y}_i \mid \text{cue}) < \text{Var}(\hat{Y}_i \mid \text{no cue}).$$

(ii) If $N^+ < N^-$, there exists a threshold Δ_k such that disagreement is non-monotone in a :

$$\text{Var}(\widehat{Y}_i \mid \text{cue}) \begin{cases} > \text{Var}(\widehat{Y}_i \mid \text{no cue}) & \text{if } a < \Delta_k, \\ = \text{Var}(\widehat{Y}_i \mid \text{no cue}) & \text{if } a = \Delta_k, \\ < \text{Var}(\widehat{Y}_i \mid \text{no cue}) & \text{if } a > \Delta_k. \end{cases}$$

Moreover, Δ_k coincides with the baseline threshold Δ ; k affects the magnitude of cue effects but not the cutoff.

Proof. With the cue, each positive signal is recalled in each of k rounds with probability $\tilde{p} = p + a$, so

$$X_{i,k}^{+,\text{cue}} \sim \text{Bin}(kN^+, \tilde{p}), \quad X_{i,k}^{-,\text{cue}} \sim \text{Bin}(kN^-, p),$$

independently. For large N^+, N^- ,

$$\begin{aligned} X_{i,k}^{+,\text{cue}} &\approx \mathcal{N}(kN^+\tilde{p}, kN^+\tilde{p}(1-\tilde{p})), \\ X_{i,k}^{-,\text{cue}} &\approx \mathcal{N}(kN^-p, kN^-p(1-p)). \end{aligned}$$

Let $Z_{i,k}^{\text{cue}} = X_{i,k}^{+,\text{cue}} - X_{i,k}^{-,\text{cue}}$. Then

$$Z_{i,k}^{\text{cue}} \sim \mathcal{N}(k(N^+\tilde{p} - N^-p), k[N^+\tilde{p}(1-\tilde{p}) + N^-p(1-p)]).$$

Therefore,

$$\tilde{q}_k = P(Z_{i,k}^{\text{cue}} > 0) = \Phi\left(\frac{k(N^+\tilde{p} - N^-p)}{\sqrt{k[N^+\tilde{p}(1-\tilde{p}) + N^-p(1-p)]}}\right) = \Phi(\sqrt{k}\tilde{c}),$$

where

$$\tilde{c} := \frac{N^+\tilde{p} - N^-p}{\sqrt{N^+\tilde{p}(1-\tilde{p}) + N^-p(1-p)}}.$$

From Proposition 11,

$$q_k = \Phi(\sqrt{k}c), \quad c := \frac{(N^+ - N^-)p}{\sqrt{Np(1-p)}}.$$

Disagreement depends on q only through

$$\text{Var}(\widehat{Y}_i) = 4q(1-q) = f(z) \quad \text{with} \quad z = \Phi^{-1}(q),$$

and f is strictly decreasing in $|z|$. Hence the cue reduces disagreement if and only if

$$|\sqrt{k}\tilde{c}| > |\sqrt{k}c| \iff |\tilde{c}| > |c|.$$

The $\sqrt{k}$ terms cancel, so the condition is *identical* to the baseline case.

If $N^+ \geq N^-$, then $\tilde{c} > c \geq 0$, implying $\tilde{q}_k > q_k > \frac{1}{2}$ and thus disagreement strictly falls.

If $N^+ < N^-$, the inequality $|\tilde{c}| > |c|$ holds if and only if $a > \Delta$, where Δ is the same cutoff as in the baseline proof:

$$\Delta = \frac{(2\lambda - 1)^2 p(1 - 2p) - 2(2\lambda - 1)p(1 - p)}{\lambda(1 - p) + (2\lambda - 1)^2 p}, \quad \lambda := \frac{N^+}{N}.$$

Therefore, disagreement increases for $a < \Delta$, decreases for $a > \Delta$, and is unchanged at $a = \Delta$. The role of k is to amplify both standardized gaps by $\sqrt{k}$, affecting magnitudes but not the threshold. $\square$

S.10 Quantitative AR(1) Belief Experiment: SMD Estimation

This appendix calibrates the imperfect-recall model in the AR(1) quantitative environment of Section 6 of the main paper using a simulation-based minimum-distance (SMD) estimator, mirroring the qualitative SMD estimation procedure in the main-paper appendix on Quantitative Estimation: Simulation Details.

Baseline model and simulation A population of DMs observes a common realized history $\{Z_1, \dots, Z_T\}$ with $T = 20$, generated from a stationary AR(1) process with persistence ρ and innovation standard deviation σ_ε . Each DM i recalls each realization independently with probability $p \in (0, 1)$ and reports the sample mean of recalled values plus a mean-zero response-noise term $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$:

$$\mu_i = \frac{\sum_{t=1}^T I_i^t Z_t}{\sum_{t=1}^T I_i^t} + \eta_i, \quad I_i^t \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p).$$

The cross-sectional disagreement in group g is $D_g = \text{Var}(\mu_i)$. Proposition A.1 in Appendix J of the main paper (the basis of Prediction 4) implies that, in expectation over recall draws, D_g scales with the realized history variance $\widehat{\text{Var}}(Z_1, \dots, Z_T)$ through the factor $(1 - p)/(Tp)$. The response-noise term σ_η^2 captures discretization, rounding, and other reporting-side noise that is orthogonal to recall and contributes additively to belief variance.

For any candidate parameter pair (p, σ_η) , we compute model-implied disagreement by Monte Carlo simulation. Specifically, we draw $R = 500$ realizations of the recall mask $I_i^{1:T} \sim \text{Bernoulli}(p)^T$ for each group's realized history, compute belief draws μ_i as above, and compute the simulated cross-sectional variance $D_g^{\text{mod}}(p, \sigma_\eta)$, which we match to the corresponding observed moment D_g^{exp} .

Exclusion of extreme outliers A material share of Day-1 belief responses lie well outside the range of the data participants were shown. These responses cannot reflect averaging over recalled values from the displayed history and are most plausibly attributable to inattention or guessing. We therefore restrict the estimation sample to Day-1 responses lying within $\pm 5 \widehat{\text{SD}}(Z)$ of the realized history's sample mean for each group. This rule retains 99.4percent of Group 1, 89.9percent of Group 2, and 100.0percent of Group 3 respondents.

Estimation We estimate $(\hat{p}, \hat{\sigma}_\eta)$ by minimizing the percent-error SMD criterion

$$(\hat{p}, \hat{\sigma}_\eta) = \arg \min_{p, \sigma_\eta} \sum_{g=1}^3 \left(\frac{D_g^{\text{mod}}(p, \sigma_\eta) - D_g^{\text{exp}}}{D_g^{\text{exp}}} \right)^2$$

over a fine grid $(p, \sigma_\eta) \in [0.02, 0.60] \times [0.00, 1.50]$, with $R = 500$ Monte Carlo draws per evaluation and a fixed random seed.

Results The SMD estimator returns $\hat{p} = 0.10$ and $\hat{\sigma}_\eta = 0.52$. Figure S.4 compares model-implied and observed Day-1 disagreement by group. The model captures the rank ordering and approximate level of D_g across groups using a single recall probability $\hat{p}$. The fit is closest for Groups 1 and 3; the residual gap for Group 2 narrows under more aggressive trimming rules.

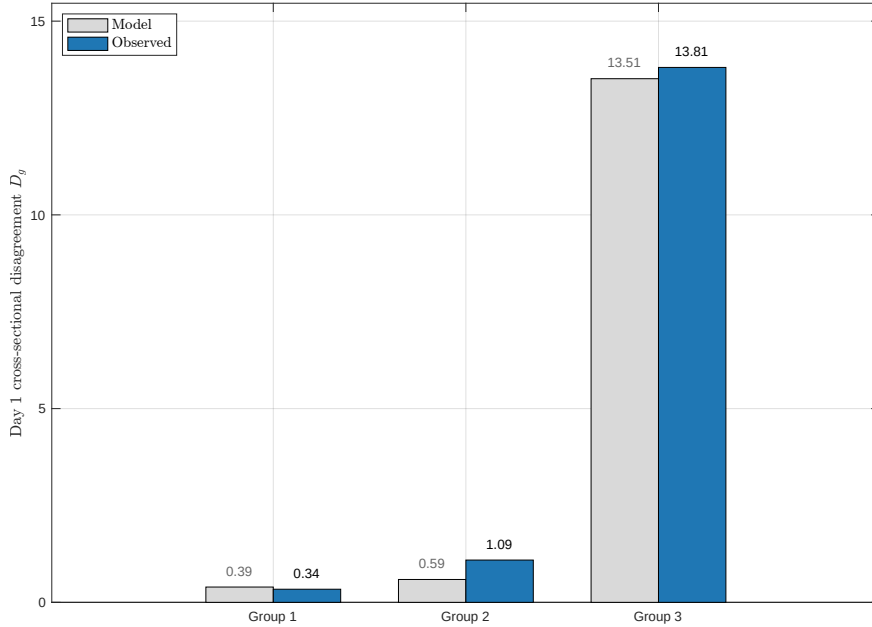


Figure S.4: AR(1) QUANTITATIVE BELIEF EXPERIMENT: SMD CALIBRATION ON DAY-1 CROSS-SECTIONAL DISAGREEMENT

Notes: The figure compares observed Day-1 cross-sectional disagreement D_g (variance of mean estimates within each group) with the model-implied disagreement at the SMD estimates $\hat{p} = 0.10$ and $\hat{\sigma}_\eta = 0.52$. The estimation sample restricts Day-1 belief responses to within $\pm 5\widehat{\text{SD}}(Z)$ of the realized history's sample mean for each group. Monte Carlo simulation uses $R = 500$ recall draws per group with a fixed random seed.

S.11 Deriving the Majority Rule from Bayesian Updating

The paper models belief formation as a two-stage process: (i) the stochastic recall of stored signals (the recall stage) and (ii) aggregation via a majority rule (equation (3.1) of the main paper) (the information-processing stage).

The majority rule can be microfounded by embedding the model in a standard Bayesian classification problem. Suppose:

1. The state is binary $Y \in \{G, B\}$ with symmetric prior

$$P(G) = P(B) = \frac{1}{2}.$$

2. Signals are i.i.d. and symmetric in diagnosticity:

$$P(s = p \mid G) = P(s = n \mid B) = b > \frac{1}{2}, \quad P(s = p \mid B) = P(s = n \mid G) = 1 - b.$$

3. Memory is imperfect: only a random subset of signals is recalled. The DM observes (X_i^+, X_i^-) , the counts of recalled positive and negative signals.

Given the recalled evidence R_i , the likelihoods are:

$$P(R_i \mid G) = b^{X_i^+} (1 - b)^{X_i^-}, \quad P(R_i \mid B) = (1 - b)^{X_i^+} b^{X_i^-}.$$

Posterior odds are therefore

$$\frac{P(G \mid R_i)}{P(B \mid R_i)} = \frac{P(G)}{P(B)} \cdot \frac{P(R_i \mid G)}{P(R_i \mid B)} = \left(\frac{b}{1 - b} \right)^{X_i^+ - X_i^-}.$$

Since $b > 1/2$, we have $\frac{b}{1-b} > 1$, so

$$P(G \mid R_i) > \frac{1}{2} \iff X_i^+ - X_i^- > 0.$$

The Bayes-optimal decision is to choose G whenever $X_i^+ > X_i^-$ and B whenever $X_i^+ < X_i^-$; at a tie ($X_i^+ = X_i^-$) the posterior equals the prior, so the DM is indifferent and any tie-breaking convention is optimal. The main paper classifies the asset as good at ties, matching the convention used in the empirical construction and in the estimation. Thus, majority voting over recalled signals, with ties broken toward G , is optimal under symmetry.

S.12 Existing Survey Responses on House Price Expectation

Case and Shiller (2003) study consumer expectations during housing booms using survey data collected in 1988 and 2003 across four U.S. metropolitan areas: Los Angeles, San Francisco, Boston, and Milwaukee. The surveys elicit respondents' expectations for house price changes over one-year and ten-year horizons. Table S.3 reproduces the exact survey questions and summarizes the distribution of responses. Across both episodes, expectations are characterized by strong optimism and remarkably little dispersion. Roughly 90 percent or more of respondents expected house prices to rise, and disagreement over both short-run and long-run price changes was minimal. Importantly,

in each survey period respondents had recently experienced sustained house price appreciation in their local markets.

Table S.3: SURVEY RESPONSES ON HOUSE PRICE EXPECTATIONS, 1988 AND 2003 (CASE AND SHILLER (2003))

<i>Question</i>	<i>Los Angeles</i>		<i>San Francisco</i>		<i>Boston</i>		<i>Milwaukee</i>	
	<i>1988</i>	<i>2003</i>	<i>1988</i>	<i>2003</i>	<i>1988</i>	<i>2003</i>	<i>1988</i>	<i>2003</i>
Do you think that housing prices in the [city] area will increase or decrease over the next several years?								
Increase	98.3	89.7	99.0	90.5	90.2	83.1	87.1	95.2
Decrease	1.7	10.3	1.0	9.5	9.8	16.9	12.9	4.8
No. of responses	240	145	199	158	194	201	233	187
How much of a change do you expect there to be in the value of your home over the next 12 months?								
Mean response (percent)	15.3	10.5	13.5	9.8	7.4	7.2	6.1	8.9
Standard error	0.8	0.6	0.6	0.6	0.6	0.4	0.5	1.0
No. of responses	217	139	185	147	176	179	217	160
On average over the next 10 years, how much do you expect the value of your property to change each year?								
Mean response (percent)	14.3	13.1	14.8	15.7	8.7	14.6	7.3	11.7
Standard error	1.2	1.2	1.4	1.8	0.6	1.8	0.5	1.3
No. of responses	208	137	181	152	177	186	211	169

Notes: This table reproduces survey results reported in Case and Shiller (2003) based on responses from homebuyers in four U.S. metropolitan areas (Los Angeles, San Francisco, Boston, and Milwaukee) in two survey waves (1988 and 2003). Respondents were asked whether they expected local housing prices to increase or decrease over the next several years, and the table reports the percentage who indicated an increase or decrease as well as the number of respondents. The table also summarizes quantitative expectations of house price changes: (i) expected percentage change in home values over the next 12 months and (ii) expected annual percentage change in home values over the next 10 years. For each question, the table reports mean expected changes (in percent), associated standard errors, and sample sizes. Data correspond to Table 9 of Case and Shiller (2003).

In the language of the model, these environments correspond to low informational ambivalence: the recent history of signals stored in memory was overwhelmingly positive. With imperfect memory, recalled information is therefore dominated by positive signals for most respondents, leading to highly aligned recalled samples across individuals. As a result, belief formation under stochastic recall naturally generates near-consensus expectations and very low disagreement.

This housing-market evidence complements the stock-market results by illustrating how environments with strongly one-sided historical experiences produce compressed belief distributions and minimal disagreement, even when recall is imperfect.

S.13 Disagreement in One-Step-Ahead Forecasts

We extend the analysis to disagreement in one-step-ahead forecasts in the AR(1) environment.

Proposition 14 (Forecast Disagreement). *For large T , conditional on the realized history $(Z_1, \dots, Z_T)$,*

$$\begin{aligned} \text{Var}\left(\widehat{Z}_{T+1,i} \mid Z_1, \dots, Z_T\right) &\approx \rho^2 p(1-p) (Z_T - \bar{Z})^2 \\ &\quad + (1 - 2\rho p + \rho^2 p) \frac{1-p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T), \end{aligned} \quad (\text{S.13.1})$$

and unconditionally,

$$\text{Var}\left(\widehat{Z}_{T+1,i}\right) \approx \rho^2 p(1-p) \text{Var}(Z_t) + (1 - 2\rho p + \rho^2 p) \frac{1-p}{Tp} \text{Var}(Z_t). \quad (\text{S.13.2})$$

Proof of Proposition 14

Fix a realized history $Z = (Z_1, \dots, Z_T)$ and define

$$\bar{Z} = \frac{1}{T} \sum_{k=1}^T Z_k.$$

All randomness below is taken with respect to DMs' recall indicators $\{I_i^k\}_{k=1}^T$.

The DM i 's one-step-ahead forecast is formed using the AR(1) structure when the most recent realization is recalled. If DM i recalls Z_T , the plug-in forecast is $(1 - \rho)\mu_i + \rho Z_T$; otherwise the DM uses μ_i as a proxy. Both cases can be written compactly as

$$\widehat{Z}_{T+1,i} = \mu_i + \rho I_i^T (Z_T - \mu_i) = (1 - \rho I_i^T) \mu_i + \rho I_i^T Z_T. \quad (\text{S.13.3})$$

Define the approximate cross-sectional mean forecast

$$m(Z) = (1 - \rho p) \bar{Z} + \rho p Z_T. \quad (\text{S.13.4})$$

This expression is obtained by replacing μ_i with $\bar{Z}$ and I_i^T with its mean p in (S.13.3). Any discrepancy between $\mathbb{E}[\widehat{Z}_{T+1,i} \mid Z]$ and $m(Z)$ is of order $O(T^{-1})$ and does not affect the leading variance terms.

Subtracting $m(Z)$ from (S.13.3) and subtracting $\bar{Z}$ yields

$$\widehat{Z}_{T+1,i} - m(Z) = (1 - \rho I_i^T)(\mu_i - \bar{Z}) + \rho(I_i^T - p)(Z_T - \bar{Z}). \quad (\text{S.13.5})$$

From Proposition A.1 in Appendix J of the main paper, conditional on Z ,

$$\text{Var}(\mu_i - \bar{Z} \mid Z) \approx \frac{1-p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T), \quad \mu_i - \bar{Z} = O_p(T^{-1/2}). \quad (\text{S.13.6})$$

Moreover, changing I_i^T perturbs both the numerator and denominator of μ_i by $O(1)$ while the denominator is $O(T)$, implying

$$\text{Cov}(I_i^T - p, \mu_i - \bar{Z} \mid Z) = O(T^{-1}). \quad (\text{S.13.7})$$

Using (S.13.5), the conditional variance decomposes as

$$\text{Var}(\widehat{Z}_{T+1,i} \mid Z) = \text{Var}((1 - \rho I_i^T)(\mu_i - \bar{Z}) \mid Z) + \text{Var}(\rho(I_i^T - p)(Z_T - \bar{Z}) \mid Z) + o(T^{-1}),$$

where the covariance term is negligible by (S.13.7).

Evaluating each component yields

$$\text{Var}(\rho(I_i^T - p)(Z_T - \bar{Z}) \mid Z) = \rho^2 p(1 - p)(Z_T - \bar{Z})^2,$$

and

$$\text{Var}((1 - \rho I_i^T)(\mu_i - \bar{Z}) \mid Z) \approx (1 - 2\rho p + \rho^2 p) \frac{1 - p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T).$$

Combining terms gives (S.13.1). Taking expectations over histories yields (S.13.2).

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